

FALCON's Response to DHL for Supply of 9K Loop CBS 5 – Bengaluru

September 16, 2025

Doc. No. - Sales/FR/PT/007/R0





Kind Attention –

M/s DHL

Offer Ref: F25-00196; Date: 16-09-2025

Subject – Techno -Commercial Offer of Loop Cross Belt Sorter 9K PPH

We are pleased to submit our Techno-Commercial Offer in response to your RFP.

As you will note, we have done an in-depth data analysis and evaluated various solution options best suited for your requirements, along with the information collected during the meetings and discussions with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to being your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Intra-logistics Automation Technologies and references.

To conclude, I would like to add my personal commitment on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

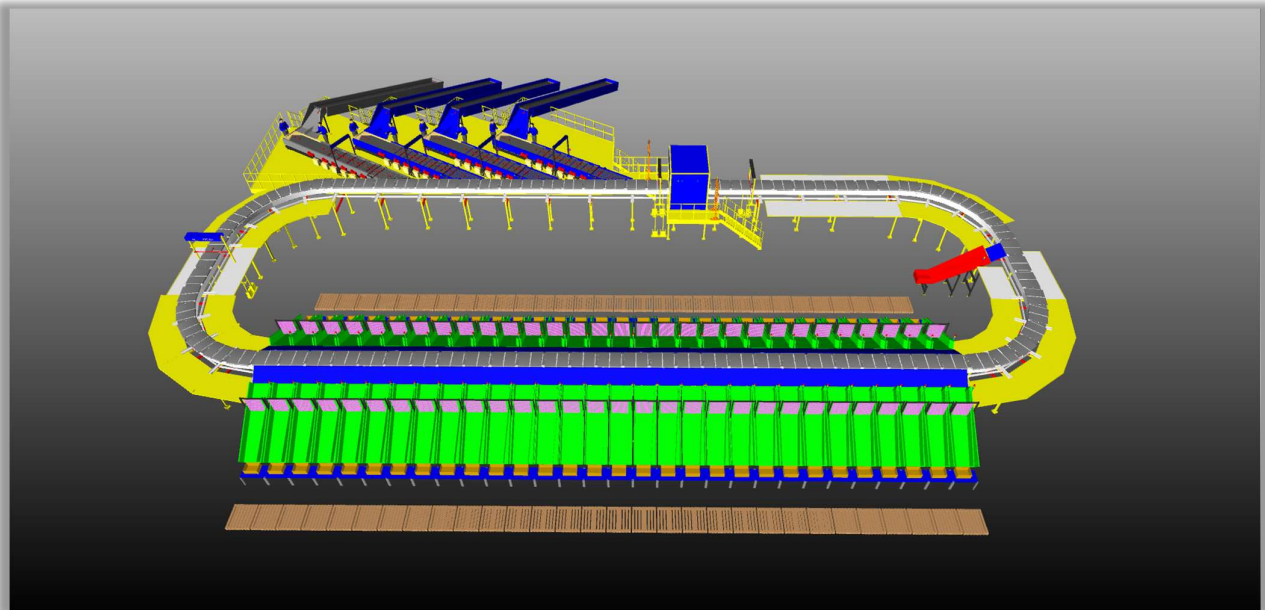
Best Regards,

Sandeep Bansal

Chief Business Officer

Response to RFP for Loop CBS 9K PPH Throughput

Proposal Reference – F25-00196



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1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Shipments Per Hour
3	ICR	Intelligent Character Recognition
4	MEZZ	Mezzanine
5	LIM	Linear Induction Motor
6	ECDS	Empty Carrier Detection System
7	AC	Alternating Current
8	DC	Direct Current
9	PLC	Programmable Logic Controller
10	IT	Information Technology
11	BOQ	Bill Of Quantity
12	I/O	Input/ Output
13	PDP	Power Distribution Panel
14	PC	Personal Computer
15	UPS	Uninterrupted Power Supply
16	CBS	Cross Belt Sorter
17	MDR	Motor Driven Roller
18	DAP	Design Approval Phase

2. Executive Summary

Falcon Autotech is pleased to confirm its great interest in responding to this DHL RFQ of Loop CBS for Bengaluru Levis facility. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening, understanding your needs, and ensuring this project's success.

Following the same objective for the Bengaluru CBS system, we are happy to offer a compliant solution meeting all technical and operational requirements at competitive price, delivering key results.

Our solution is based on the following key characteristics:

Shipment Sorter 9K PPH:-

- *One Single deck loop of shipment sorter, employing Falcon's cross-belt technology, have been designed to handle a throughput of approximately 9K packages per hour presently & scope to upgrade upto 12K PPH in future.*
- *Falcon's unique Launch pad system connecting induct lines with bulk shipment*
- *There is one induction zones featuring three induct lines and provision for One extra line in future.*
- *The system has 60 Direct chutes and one rejection chutes (L-Type).*

This solution has been crafted especially keeping DHL's technical and operational requirements, as listed in the face to face meetings, making it a tailor-made system delivering a faster TAT and an efficient material flow.

*The Falcon Autotech Cross Belt Sorter, proudly **MADE IN INDIA**, incorporates in-house designed and developed system components and associated peripherals.*

1. Falcon's reliable Shipment sortation systems

Falcon Autotech has over 65+ Cross belt sorter system installed globally which are used by most innovative brands such as Shadowfax, Flipkart, Amazon, Delhivery, Asendia, Fastway and many more. The main and critical components of the Falcon Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from industry leading suppliers in the world, such as SEW, Siemens, SICK, Vahle, Forbo etc. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. Commitment to quality systems

Demonstrating Falcon's clear commitment to the DHL satisfaction, the shipment sortation system, parts and services will be under warranty for 12 months from installation go-live.

3. A strong and already successful partnership between DHL & Falcon

DHL has been Falcon's customer since long and using multiple Falcon Systems. For each of these projects, several advanced tools to improve the service's quality and efficiency are provided: Service Desk, Hotline support, remote access, regular governance meetings and monthly reports.

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0

Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5

Product Lines

15+

Countries with Live Installations

1800+

Total Installed Systems Globally

1Million+

Sorts Per Hour

600+

Employees

Long-standing Relationships with industry leaders

Ecom Retail

Flipkart

amazon

Lazada

نون

CEP

aramex

BLUE DART

DELHIVERY

Ecom Express

Distribution & Others

G

SWIGGY

TITAN

Wendlandt (Schneider Electric)

News & Articles

FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

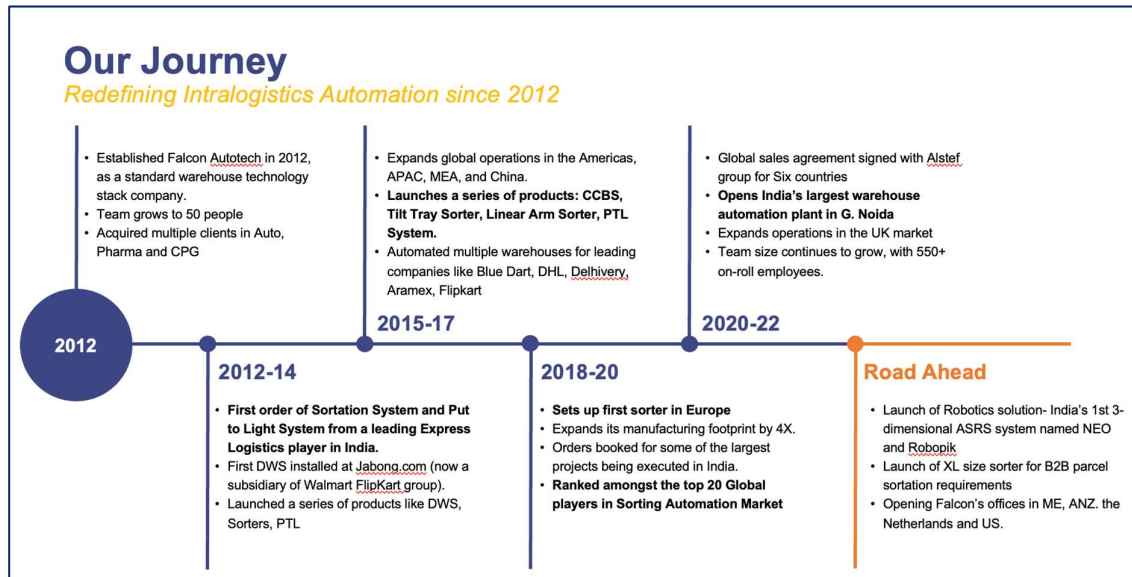
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Product and Solutions

With 100% focused on Parcels, Packages, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- PopUp Sorter
- Sweep Sorter
- Pusher Sorter



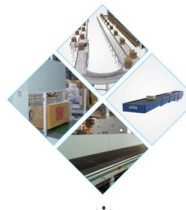
PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
- R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
- Mini (600MM)
- Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

FACIS

SORT IT

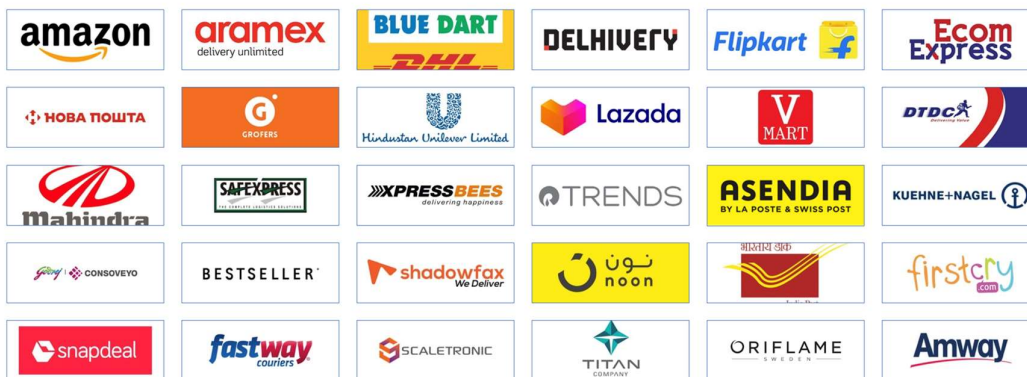
Control IT

Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

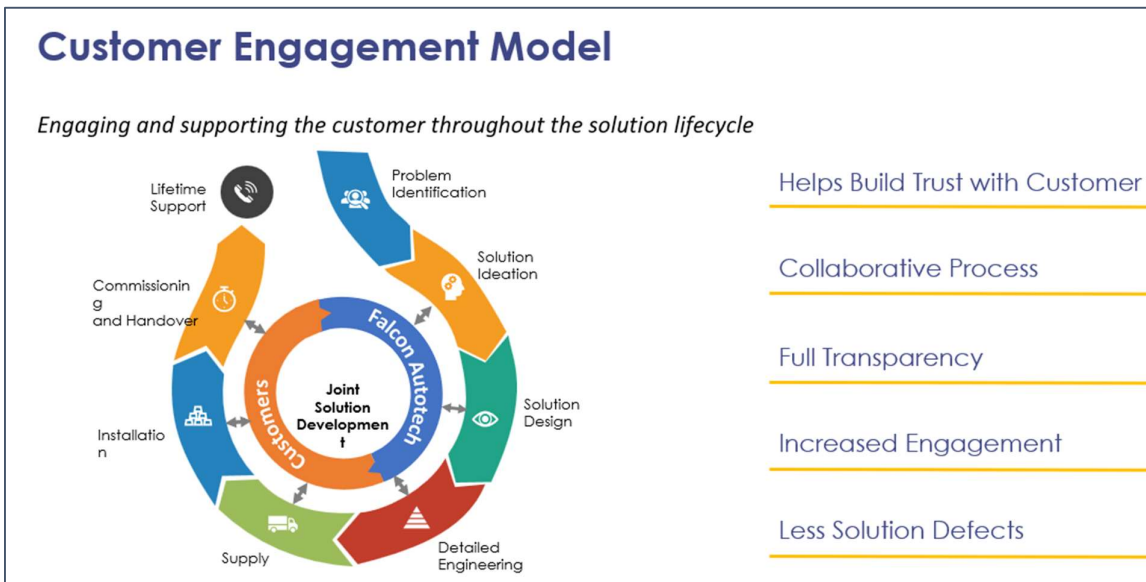
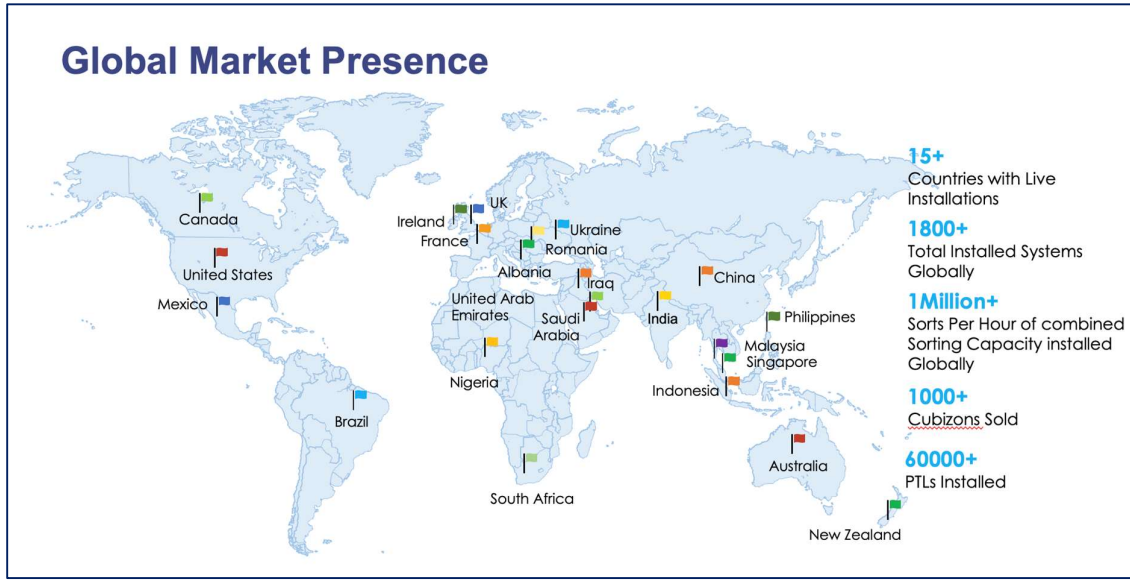
Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

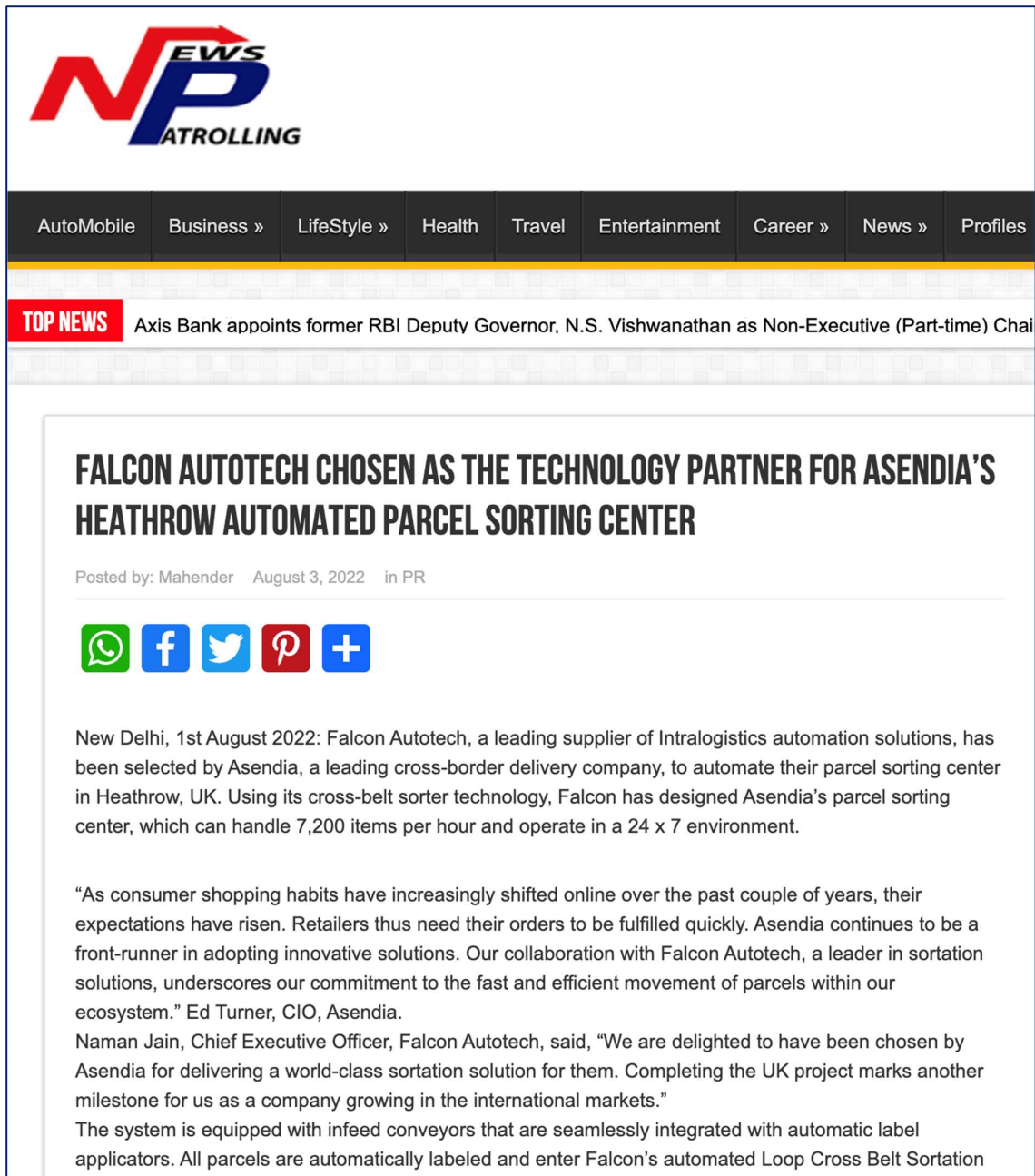


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



NP NEWS PATROLLING

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TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chair

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation



Breaking Barriers and Optimizing Efficiency – Journey of Delivering India's Largest Sortation System

"It was a bold decision. If it didn't work, our company would have shut down," says Naman Jain, founder and CEO of warehouse automation startup Falcon Autotech, referring to a big gamble he had taken more than a decade ago. The year was 2013, Jain and his team were negotiating a deal with a major e-commerce logistics company that had just placed an updated order for a sorter with a capacity of 6,000 parcels per hour. That was 4x the specification earlier agreed upon.

"All the founders came together, and we took a calculated risk," recalls Jain. At the time, Falcon did not have the technology to develop such sorters, and hence, the said client refused to pay any money upfront and offered to pay the machine's price in monthly installments. However, there was a caveat: if it failed even once, Falcon would simply take the machine back and refund the entire amount. Falcon's 'calculated risk' paid off. The machine worked flawlessly and was handed over to the parcel company after 36 months. "We are now doing projects worth INR100 crore. That was unthinkable 10 years back," says Jain.



Economic Times Features Falcon Autotech

Naman Jain
Chief Executive Officer

From sorters to conveyors & robot based systems, here's the top tech warehouses are investing in

India's warehousing industry has travelled a long distance from "godowns" to modern storage facilities called Grade A warehouses. Warehouse automation is a gradual process. India started off late, but that may actually help it bypass some of the mid-age technologies and adopt the latest ones. While several players are still warming up to automation, which are the



Falcon Announces Successful Go-Live of Automated Parcel Sortation Solution at DTDC's Chennai Facility

New Delhi, India – 2nd Aug 2023: Falcon Autotech, a leading supplier of intralogistics automation solutions, has been selected by DTDC Express Ltd, one of India's leading integrated express logistics company, to automate its parcel sorting operations at its super hub of 1,75,000 sq ft in Chennai, Tamil Nadu. Using its cross belt sorter technology, Falcon has designed DTDC's parcel sorting system, which can handle 6,000 parcels per hour, operate in a 24 X 7 environment, and can be expanded to cater to future growth.

The new linear cross-belt solution leverages cutting-edge technology to automate key sorting processes in DTDC's warehouse, including parcel profiling and sorting. This solution is designed to optimize the space requirements for sorting operations, increase efficiency, and reduce operational costs.



Transforming Indian Retail: The Impact of ASRS on Warehousing Operations

In recent years, the Indian retail industry has experienced unprecedented growth, propelling the nation to become the fifth-largest global retail destination. The industry, marked by its resilience and adaptability, went through significant transformations during the pandemic. While online shopping saw a remarkable surge, the reopening of physical stores ushered in a resurgence of the multi-sensory shopping experience. As a testament to this growth, shopping malls now encompass an astonishing 23.25 million square feet of retail space. However, the evolving consumer preferences have placed substantial pressure on businesses, necessitating the seamless integration of e-commerce and in-store experiences, which, in turn, has led to complex logistics challenges. The traditional warehousing systems struggled to cope with the demand for faster processing and the requirement for expanded storage capacities.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

5.1 Project 1- (CEP Client, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

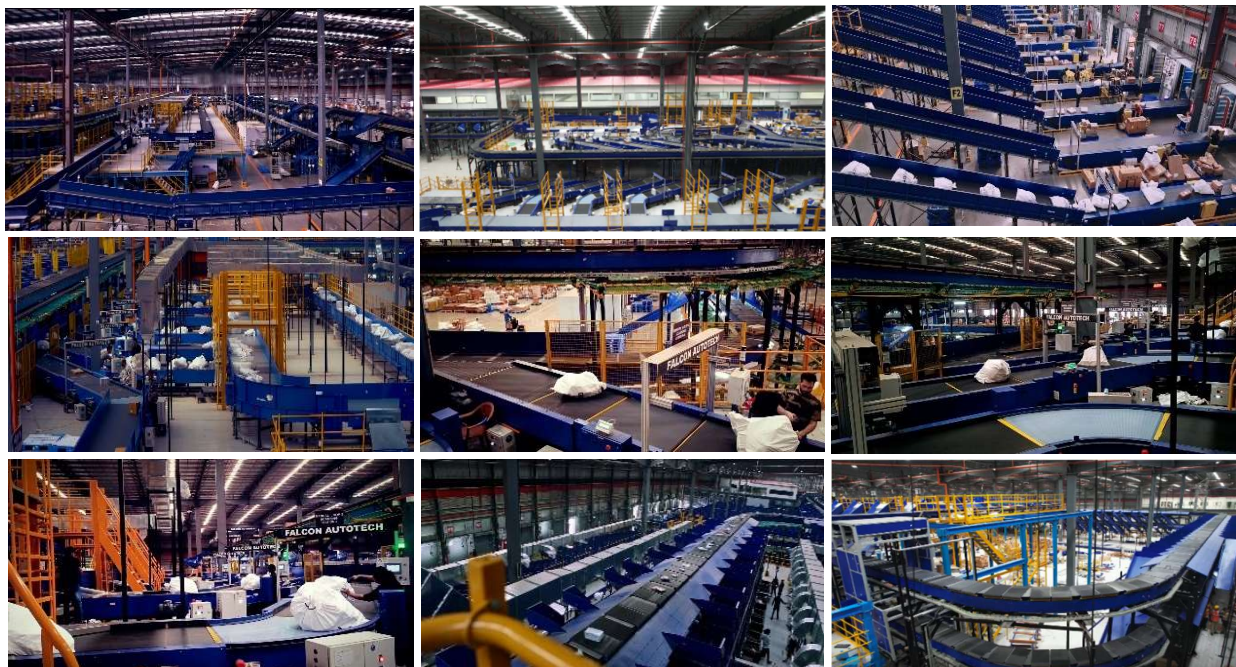
Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- | | |
|---------------------------------------|--------------------------------------|
| • 2 Sets of Double Decker CBS Sorters | • Spiral Chutes with Braking rollers |
| • Mezzanine Structures | • 5-Sided Scanning Tunnels |
| • Automated Singulators | • High speed weighing conveyors |
| • Fully Automatic Inductions | • Direct Bagging Chutes |
| • Semi-Automatic Inductions | • Put to Light Chutes |
| • Telescopic belt conveyors | • Volume Distribution systems |
| • PVC Belt Conveyors | • High Availability Server Systems |
| • Modular Belt Conveyors | • WCS |

Site Pictures -



5.2 Project 2- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

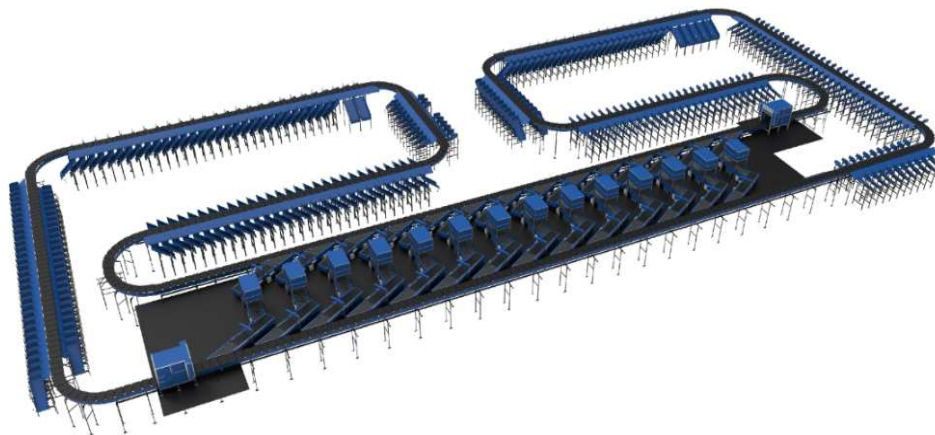
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



5.3 Project 3- (Ecommerce Client, India)

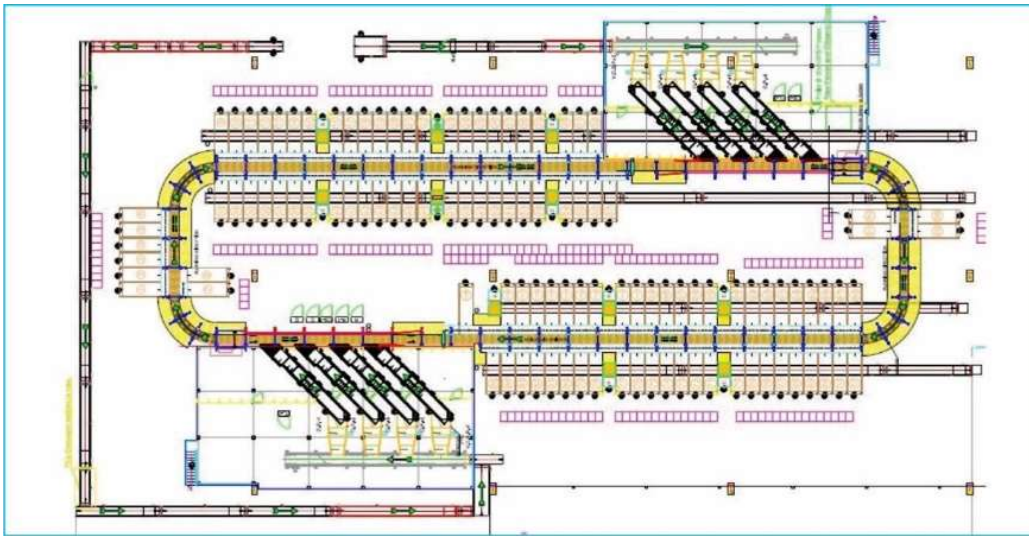
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



5.4 Project 4- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

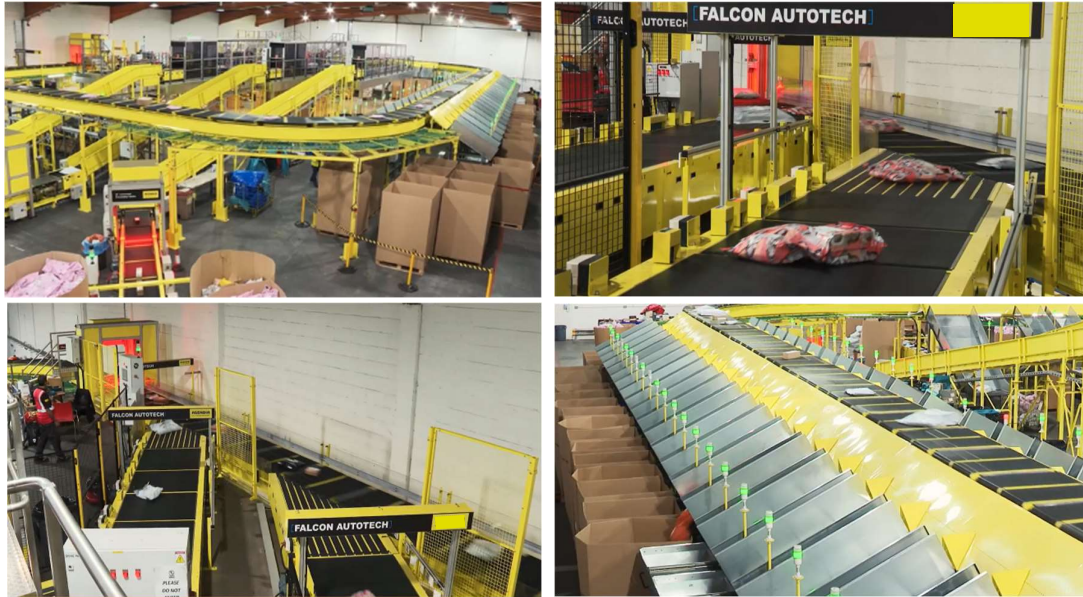
Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

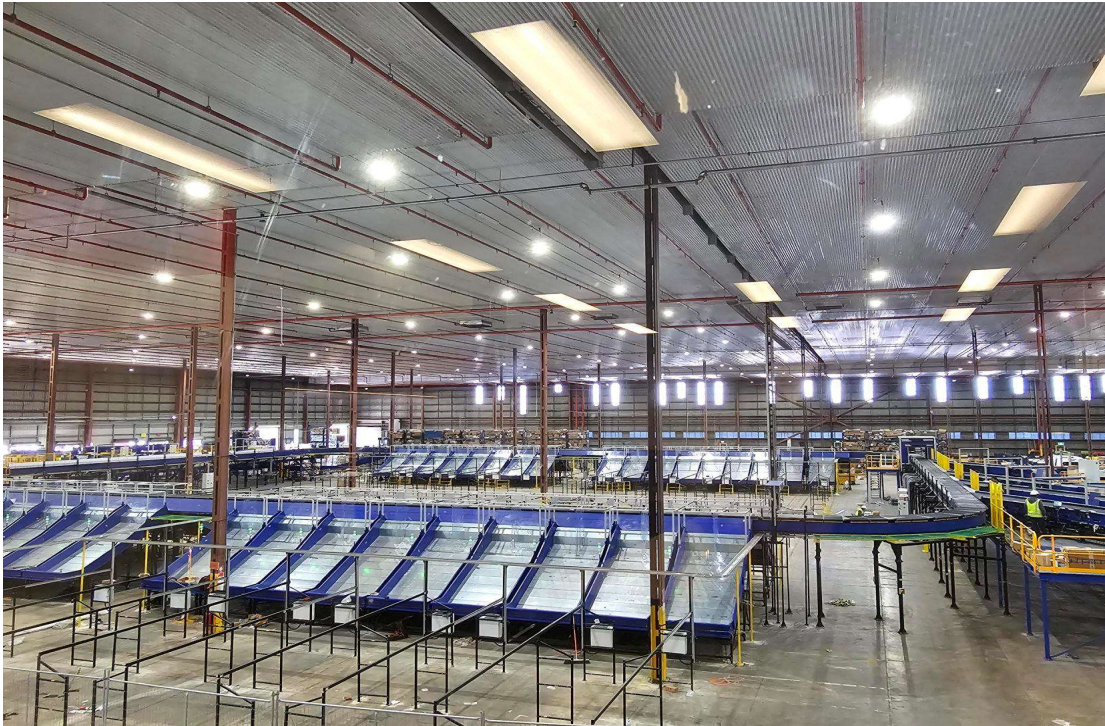
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



6. Handled Shipment Spectrum

As per the shipment spectrum data provided in the RFP documents, Falcon has studied and analysed the shipment spectrum in detail.

Falcon purposes to use its “Dual Belt Regular Configuration” for shipment sorter to provide the maximum benefits to DHL in-terms of handling various sizes and weight.

6.1 Shipment (Non-Large) size loadable on the shipment sorter

Falcon's Dual belt regular Cross belt sorter has a capability to handle the below mentioned shipment sizes and weight.

Specification	Unit	Value
Max Length	mm	600
Max Width	mm	400
Max Height	mm	300
Max Weight	Kg	10
Min length	mm	100
Min Width	mm	100
Min Height	mm	5
Min Weight	gm	100

6.2 Shipments to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of shipment cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Shipments shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The shipments must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.3 Shipment not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile shipments with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6.4 Pictures of Sortable Shipments

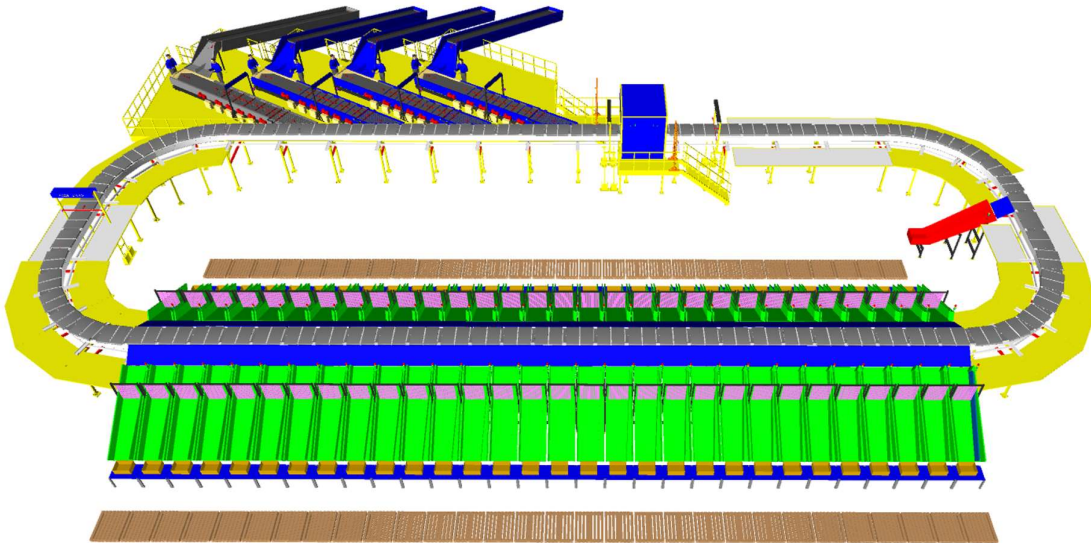
PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre-Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre-Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often Import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Flats	A parcel that often contains documents.
	Totes	A large plastic, lidded box.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding several other parcels within it.

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Loop CBS for sorting shipment & Linear CBS for sorting Bags & Semi-Large Shipments as per DHL's requirements.

7.2 Summary of the System



Process flow of the Shipment Loop CBS System-

1. **Infeed System:** - Bags containing shipments are unsealed and dumped in bulk onto the infeed Launchpads of the Cross belt sorter equipped inclined foot operated switch to pull up material as per requirement. The shipments ascend to a higher level and arrive at the induct zone mezz.
2. **Inducts:** - After the shipments are collected in the launchpad chute, an operator picks and positions each shipment on the induct line, ensuring that the shipment is properly aligned and that its barcode is facing upwards. The feedlines then automatically induct the shipments onto the Cross-Belt Sorter Loop.
3. **Loop CBS:** - Once the shipments have entered the main loop, the Cross-Belt Sorter (CBS) efficiently sorts the shipments into their respective output chutes by utilizing the data provided by DHL's sorting logic.
4. **Output Chutes:** - The shipments are discharged into two types of chutes.
 - a. **Direct Chutes (L-type)** - Within a loop CBS system, there are a total of 60 straight direct chutes. Shipments collected within these chutes are directly loaded into outbound store boxes.
 - b. **Rejection Chute-** One rejection chutes is present to handle rejected Shipments.

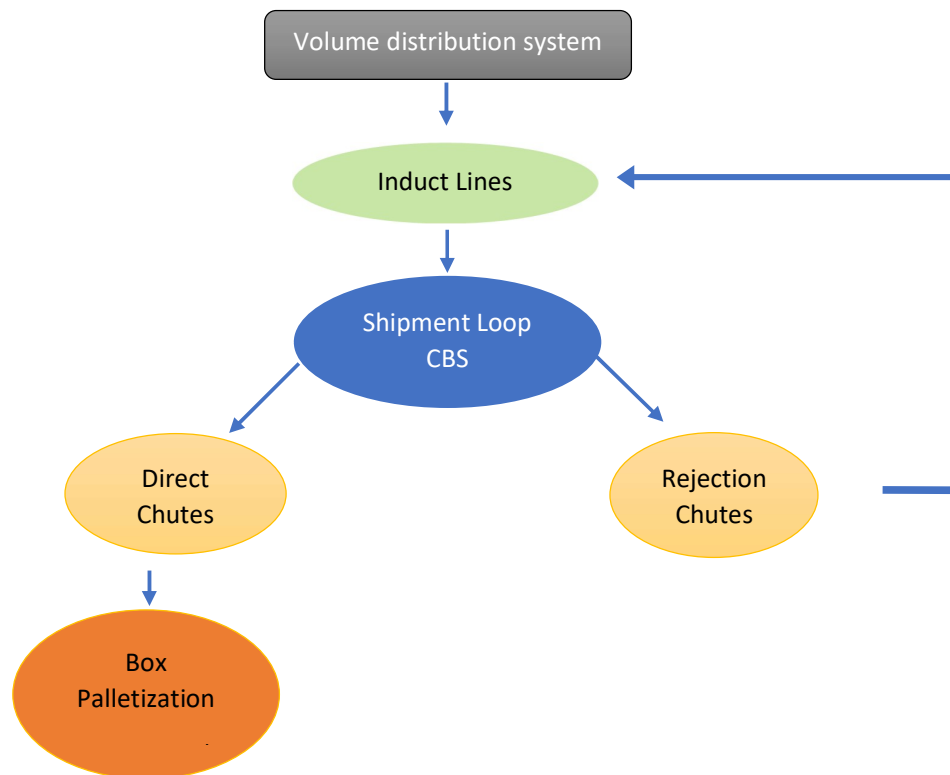
7.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

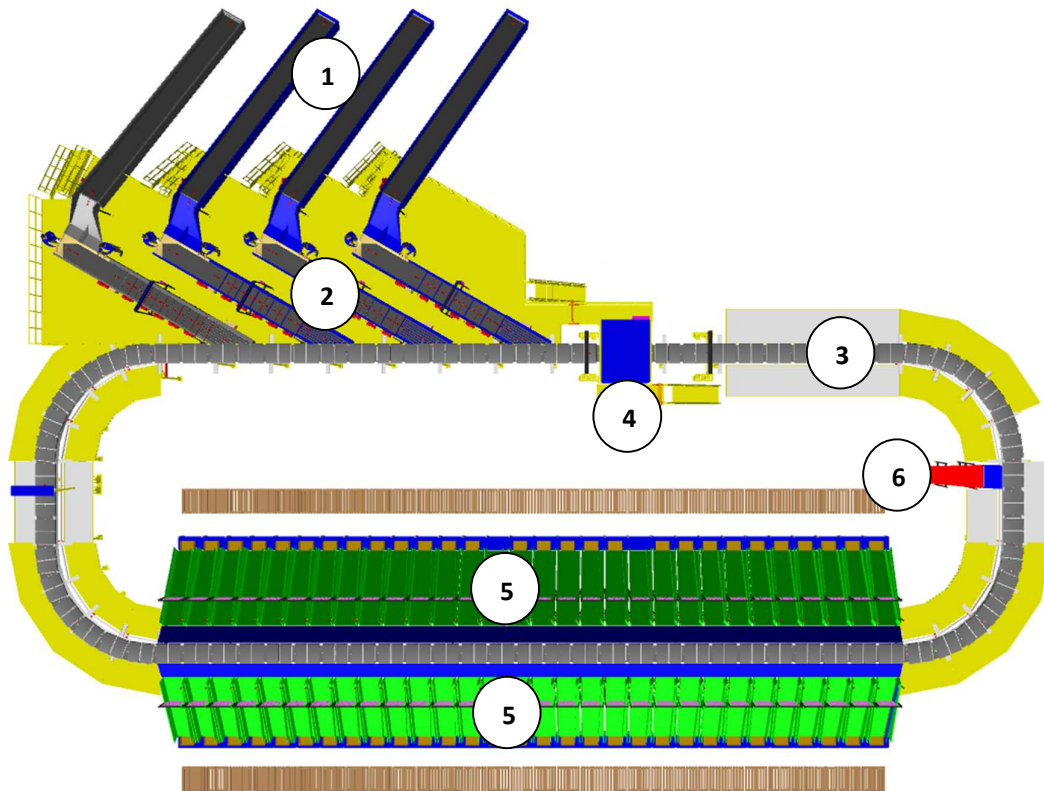
1. High operational throughput
2. Improved productivity
3. Ability to serve entire range of conveyable Shipments with a sorter's speed of up to 2.1 m/s for shipment sorter.
4. Low occupancy of floor space in the building.
5. Narrow discharge centers for the increased number of splits in limited space.
6. Designed for minimal upkeep, ensuring consistent reliability and ease of support.
7. Life Cycle Value.
8. FALCON's CBS can adapt to changing business requirements by adjusting its speed. to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

8. Concept Description

8.1 Shipment Sorter



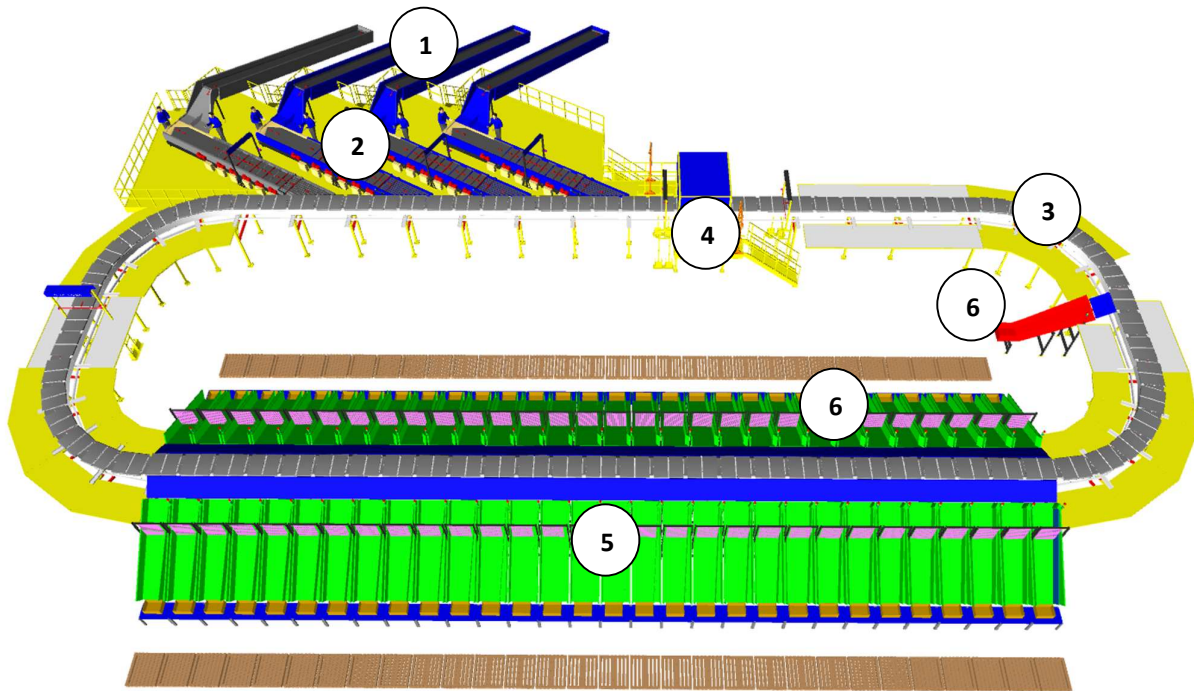
9. Layout Overview



Legend

- 1- Launchpads
- 2- Induct lines
- 3- Main Loop CBS
- 4- Scanning Tunnel
- 5- Direct chutes
- 6- Rejection chutes

10. Layout View of Shipment Sorter



Legend

- 1- Launchpads
- 2- Induct lines
- 3- Main Loop CBS
- 4- Scanning Tunnel
- 5- Direct chutes
- 6- Rejection chutes

11. Proposed System Capacity Calculations

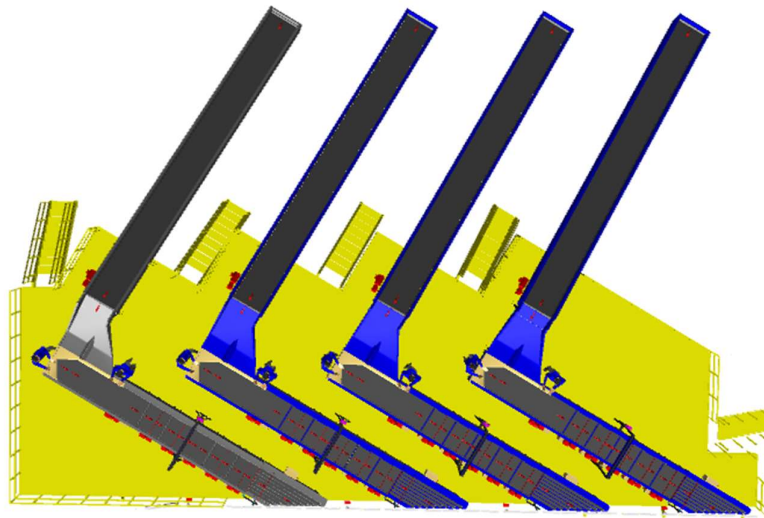
11.1 Sorter System Capacity

The following table shows the throughput calculation for the sortation system designed based on DHL requirements.

SPECIFICATION	VALUE
Sorter Type	Loop CBS
Speed	2.1 m/s
Pitch	1175 mm
Carrier per hour	6,434 CPH
Belts per hour	12,868 BPH
Effective Designed Throughput of loop CBS (A)	12,868 Shipments per hour
Per Feedline Capacity	3,000 Shipments per hour
No of Feedlines both zones (Each Deck)	3 Nos.
Total Feedline designed capacity (B)	9,000 Shipments per hour
System designed throughput (Min of A&B)	9,000 Shipments per hour

12. System description

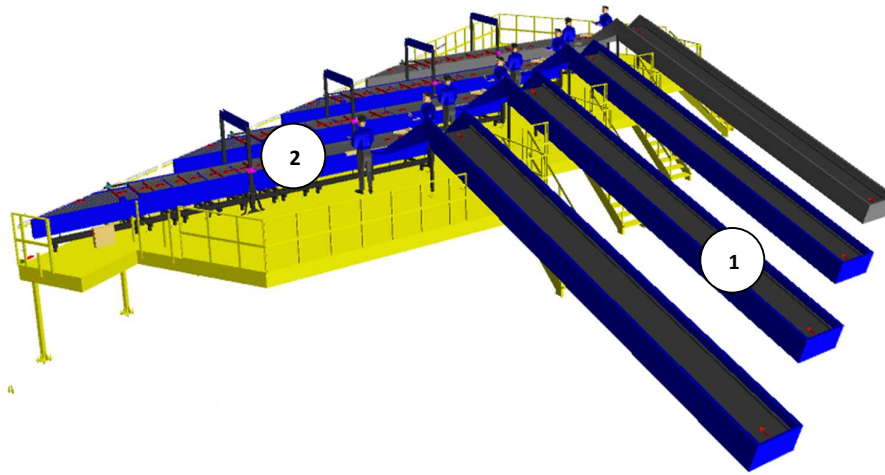
12.1 Infeed System Shipment sorter



The layout showcases a comprehensive arrangement of conveyors that function as infeed conveyors for the shipment sorter system. These conveyors are strategically designed to ensure efficient and seamless handling of shipments as they enter the induct zone.

The 3K configuration of the induct lines maximizes the handling capacity within each induct line, ensuring that high volumes of shipments can be managed simultaneously. This design minimizes bottlenecks and enhances the overall throughput of the sorting operation.

12.2 Infeed System Description



Legend

- 1- Launchpad
- 2- Induct Line

12.2.1 Infeed System

Infeed conveyor system consists of below components:

1. Inclined PVC Conveyor
2. Launchpad chutes Chute

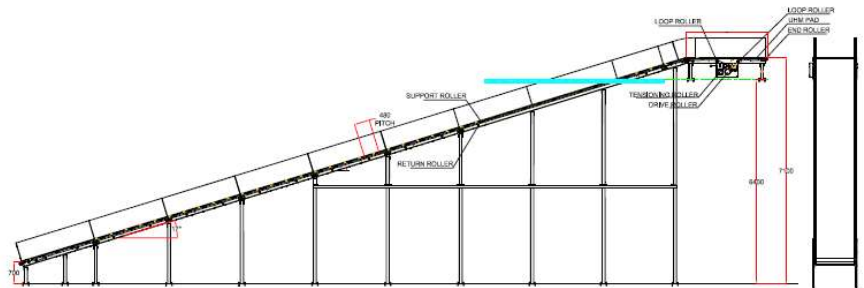
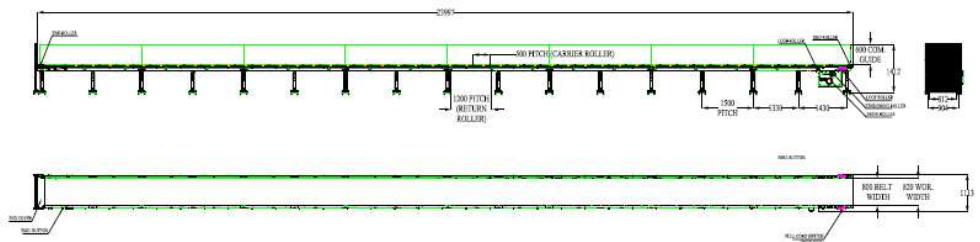
12.2.1.1 Inclined Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

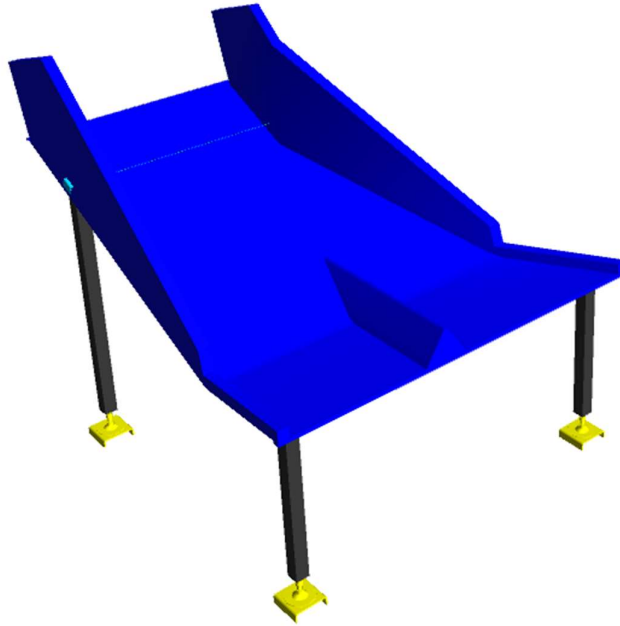
- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1200
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradley
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

12.2.1.2 Launchpad Chutes Chutes

A LP chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level (at an operator level). It utilizes base metal to guide and support the flow of items along the chute & collect the shipments within its area.



Calculation for Design: We have considered an average parcel size of 350 X 250 X 250 mm while designing the capacity of chutes.

Average parcel size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.35	0.25	0.25	0.021
Collection type Chute	2	1.7	0.4	1.36

Assuming a **60%** occupancy rate, about 38 **parcels** will readily fit into these chutes.

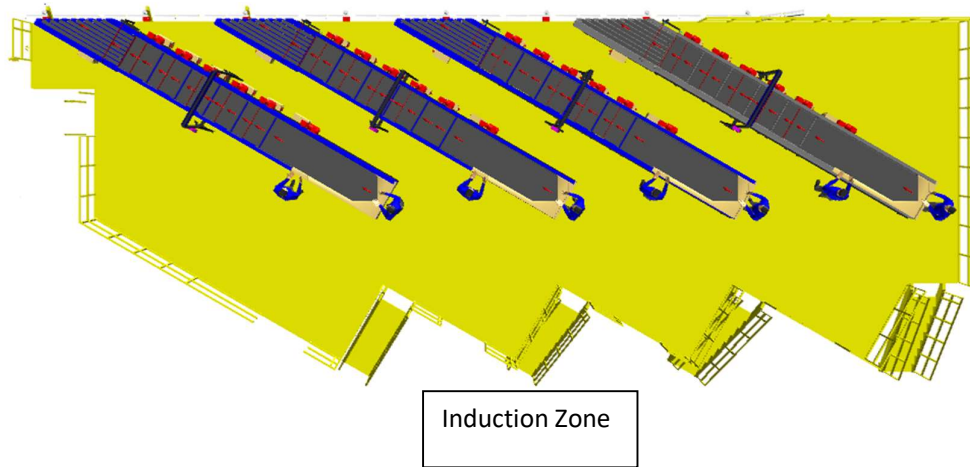
Each Chute is equipped with below components to ease the operations:

- Foot mounted Accelerator

12.3 Induction to Sorter: Feedlines

The main purpose of the feedlines is to transport the shipments from the VDS chutes to the sorter by means Loading conveyors, Weighing, Buffer/Spacing Conveyor and Angle Merge conveyors. The flow of shipments on the Inducts are controlled by sensors in such a way that the required gap is ensured between the shipments.

These Feedlines are designed especially for DHL Group considering Operator Ease and ergonomics. Where the operator ensures proper placement of the shipments on the Induct.



12.3.1 Feedlines

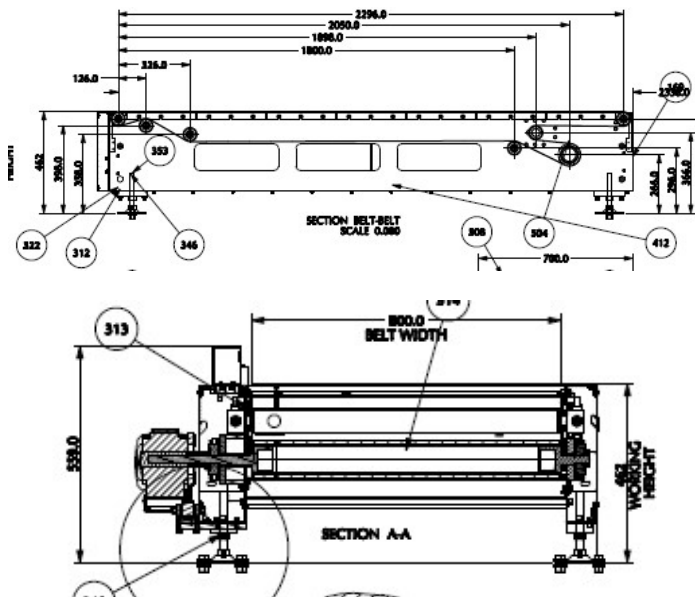
Loop CBS has a total 03 Automatic Feedlines with Designed IPP of 3000 PPH.

1. Each Feedline has Four Conveyor Modules-
 - Orientation/Loading Conveyor- 1 Nos.
 - Buffer Conveyor- 5 Nos.
 - Intelligent Merge- 1 Nos.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.
3. Over the feedline, the shipments are evenly spaced and smoothly transferred to the main Loop.

12.3.1.1 Buffer Conveyors

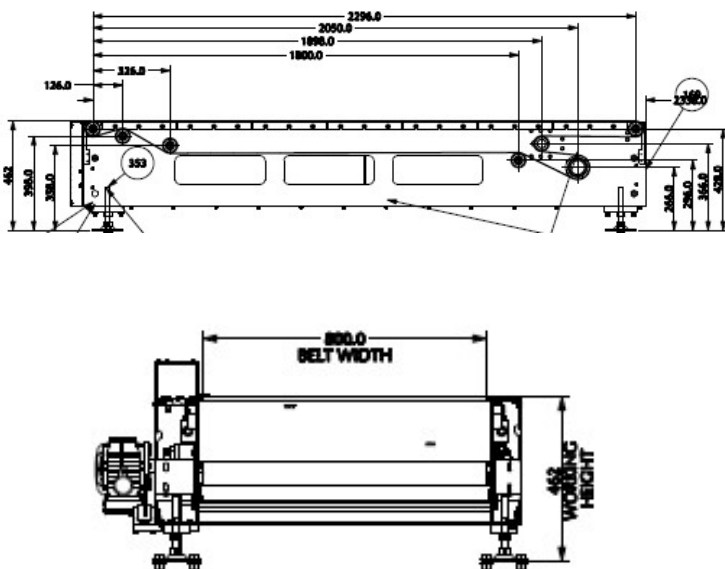
A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.



12.3.1.2 Orientation Conveyor

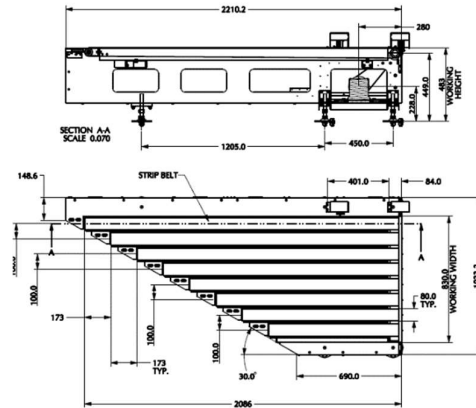
An Orientation conveyor is a type of conveyor system used to align & Release the products for induction onto CBS. It serves as the initial point of entry where products are collected and conveyed to subsequent stages of the process.



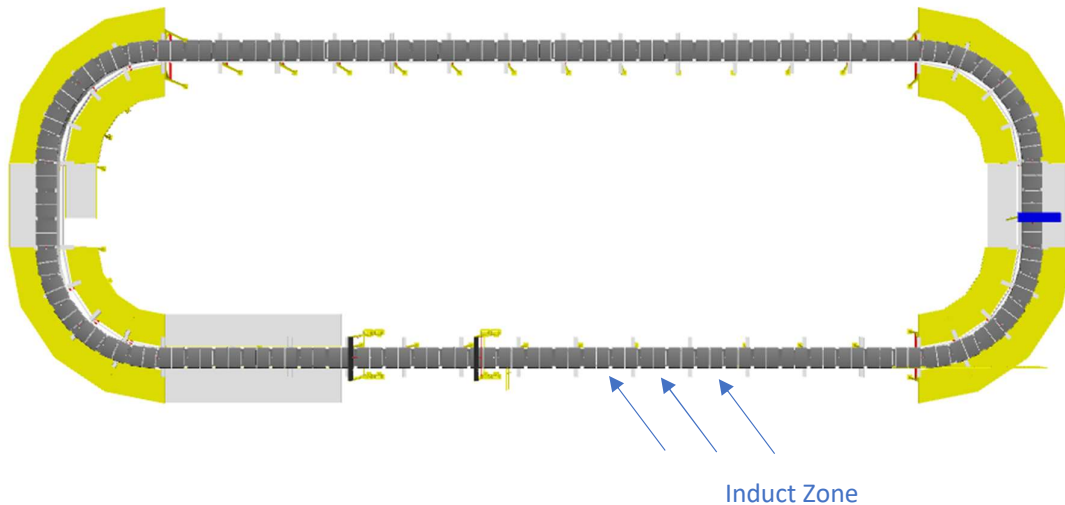
12.3.1.3 Intelligent Merge Conveyor

An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular high-speed conveyor used for inducing shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.



12.3.2 Main Loop



1. The proposed Single Deck CBS Sorter will be installed on ground level. The Sorter runs at 3000 mm from ground level.
2. We have provided loop with Dual Belt Carriers with 1175 mm pitch and Belt Size of 495 x 800mm.
3. The loop CBS has a total length of 70 m & 120 number of carriers.
4. As Shipments pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches the respective chute, the carrier carrying the shipment for the chute will actuate and drop the shipment in the assigned chute.

12.3.3 Sorting Output

12.3.3.1 Direct Bagging Chute

A Direct chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & collect the shipments into bags/boxes placed at the end of each chute.

Each Direct bagging chute has capacity to accumulate 33 shipments, The sorted shipments will travel along these collection chutes and get accumulated at the bottom of the chute.

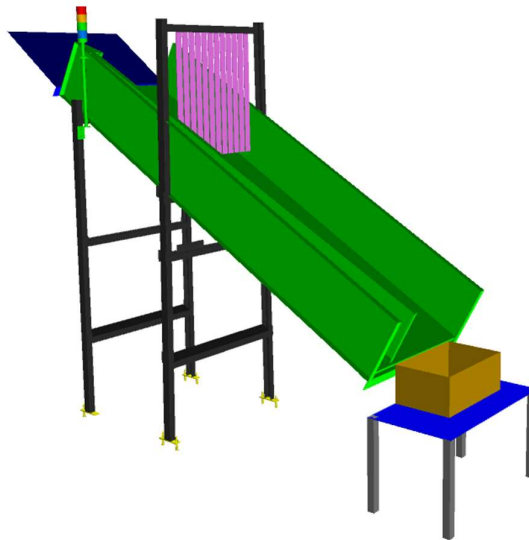


Image for representation purpose only

Calculation for Design: We have considered an average shipment size of 350 X 250 X 250 mm while designing the capacity of chutes.

Average shipment size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.25	0.25	0.35	0.022
Collection type Chute	3.4	0.9	0.4	1.22

Assuming a 60% occupancy rate, about **33 shipments** will readily fit into these chutes. 67

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12.3.3.2 Rejection Chute (L-Type)

Rejection Chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & end gate can be closed to collect the shipments within its area.

Rejection chute has capacity to accumulate 33 shipments, The sorted shipments will directly deposit into roll cage trolley place below.



Image for representation purpose only

Calculation for Design: We have considered an average shipment size of 350 X 250 X 250 mm while designing the capacity of chutes.

Average shipment size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.3	0.3	0.3	0.027
Collection type PTL Chute	3.4	0.9	0.4	1.22

Assuming a 60% occupancy rate, about **33 shipments** will readily fit into these chutes.

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Tower Light

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Tower Lamp	To indicate the chute status

13. Description of Components of Equipment

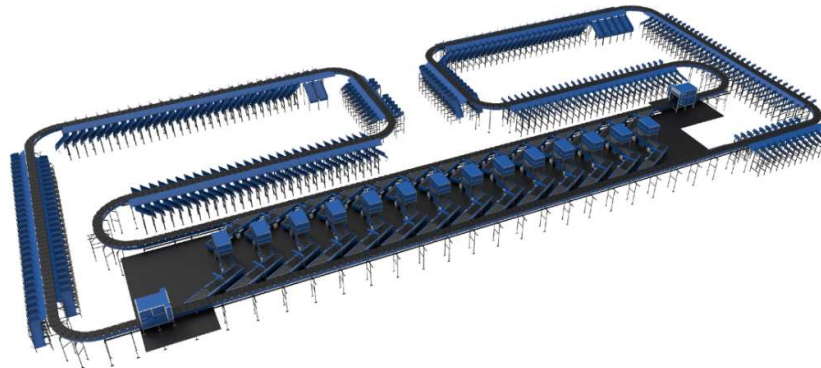
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

13.1 Cross Belt Sorter -

Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



13.1.1 CBS Carrier

Falcon's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners" and non-sortable shipment recirculation.



13.1.2 Servo Roller

High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



13.1.3 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminum which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



13.1.4 Carrier Wheels

To enhance the reliability of CBS, we will use reputed make Wheels in Non-Large CBS. Same wheels are used in Falcon XL CBS, where pay load capacity of each carrier is 40 kg as compared to Non-Large CBS, where pay load capacity is 10 kg per carrier.



Specs./ Make	Existing Make	Faigle
Wheel Dia	75 mm	100 mm
Revolutions	X	1.33X
Load bearing capacity	150 Kg	175 Kg

13.1.5 Non-contact based Linear motor drive

Extremely High Energy Efficient Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.



13.1.6 Power Transmission

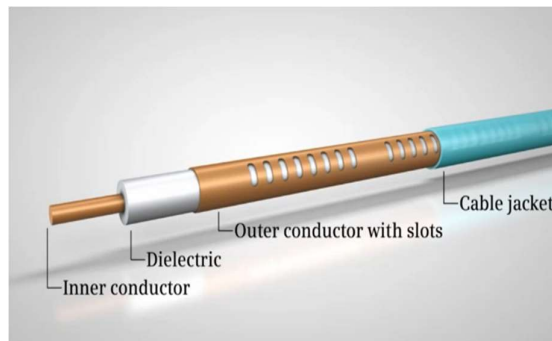
Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



13.1.7 Data Transmission

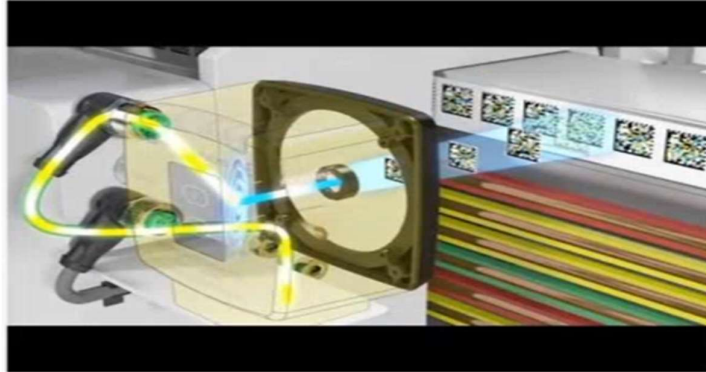
The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.

Leaky wave cable position



13.1.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



13.1.9 Technical specifications of Sorter (Shipment Sorter)

Items	Make
Sorter Carrier Type	Loop CBS
Sorter Speed in m/s	Up to 2.1 m/s
Sorter Loop Length	~70 m
Sorter Actuation Technology	Electric
Chutes	
Direct chute	60
Irregular chutes	24
Rejection Chutes	1
Height of Sorter	~3000 mm
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
No. of Carriers (Dual belt carriers)	120 Nos.
Drives	
Motor/Drive type	LIM
Power Consumption	Will be specified during DAP
Feedlines	3 Nos
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	04 Nos
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

13.2 Scanning & Sensing on Main Sorter Loop

13.2.1 Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan shipment 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time. As per DHL requirements, Scanning is in DHL scope.

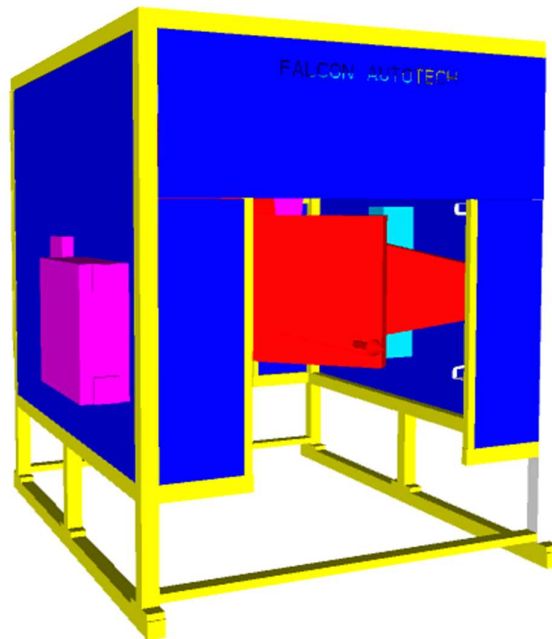
Falcon needs to supply Volume Measurement System. VMS 4200 quoted as option for Dimensioning of Shipments.



13.2.2 Automatic Barcode Scanning & Dimensioning System

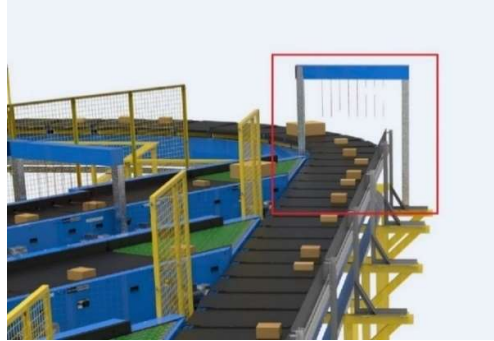
Top Sided Barcode Scanner present on CBS loop to scan 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time. As per DHL requirements, Scanning is in DHL scope.

Falcon needs to supply Volume Measurement System. VMS 4200 quoted for Dimensioning of Shipments.



13.2.3 Shipment Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple shipment positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Shipment on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



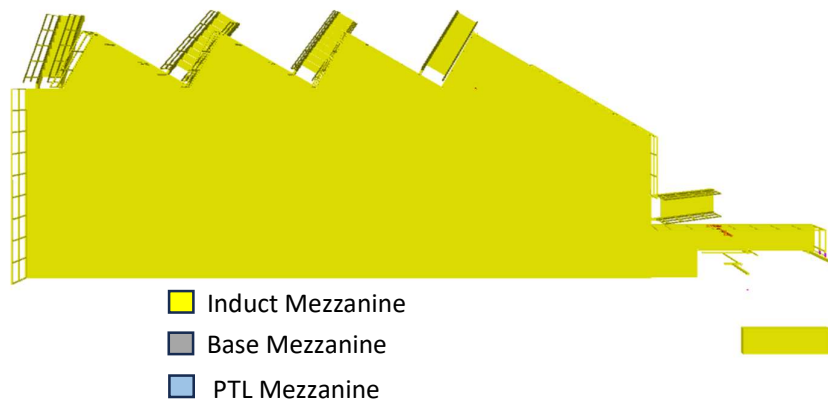
13.2.4 Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares the Feedlines to load the shipments on the empty carriers. (Camera based)



13.3 Steel Works – Mezzanine, Maintenance Platforms & Staircases-

13.3.1 Mezzanine



- Total Induct Mezzanine Area: 142 Sq. Metres
- No of Stairs: 04 Nos

14. Proposed System Technical Details (Shipment Sorter)

14.1 Mechanical equipment

Pos.	Qty.	Description	Value
1	1	Infeed System Launchpad Infeed Lines	03 Nos
2	3	Feedlines Consists of - Receiving Conveyor - Buffer Conveyor - Intelligent merge - MLG	1 Set 5 Set 1 Set 1 Set
4	1	Sorter Loop Cross Belt Sorter - Sorter Height - Sorter Length - Sorter Speed - Sorter Drive - Carrier Pitch Including: - Standard Sorter Supports - Empty Carrier Detection System - Product Centring System - Dimension Scanning System - E-Stops - Hooters - Fencing	3000 m ~70 m ~2.1 m/s Linear Induction Motor 1175 mm
5	1	Sorter Outputs - Direct Dump chutes - Rejection chutes	60 Nos 104 Nos
8	1	Steel Works - Operator and VDS Platform - Stairs	~142 SQM 04 Nos

14.2 Electrical Equipment

Electricals			
1	1	Consists of Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

14.3 Control System

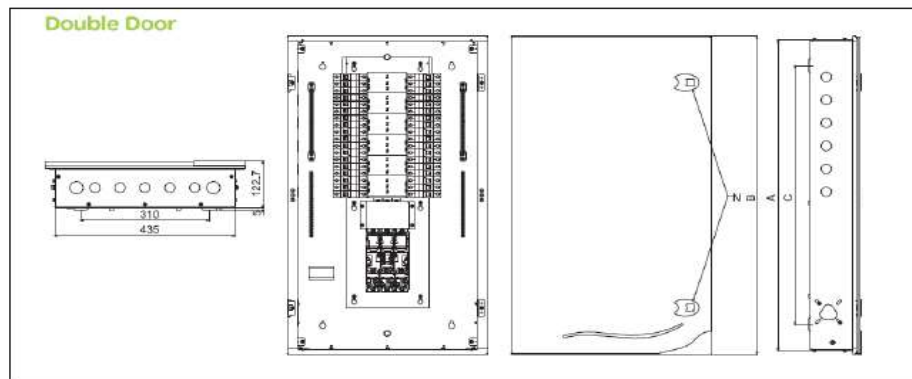
Components			
1	1	Consists of Siemen's PLC based Control System with SCADA Industrial Switch	1 Nos. As per requirement

15. Electrical System

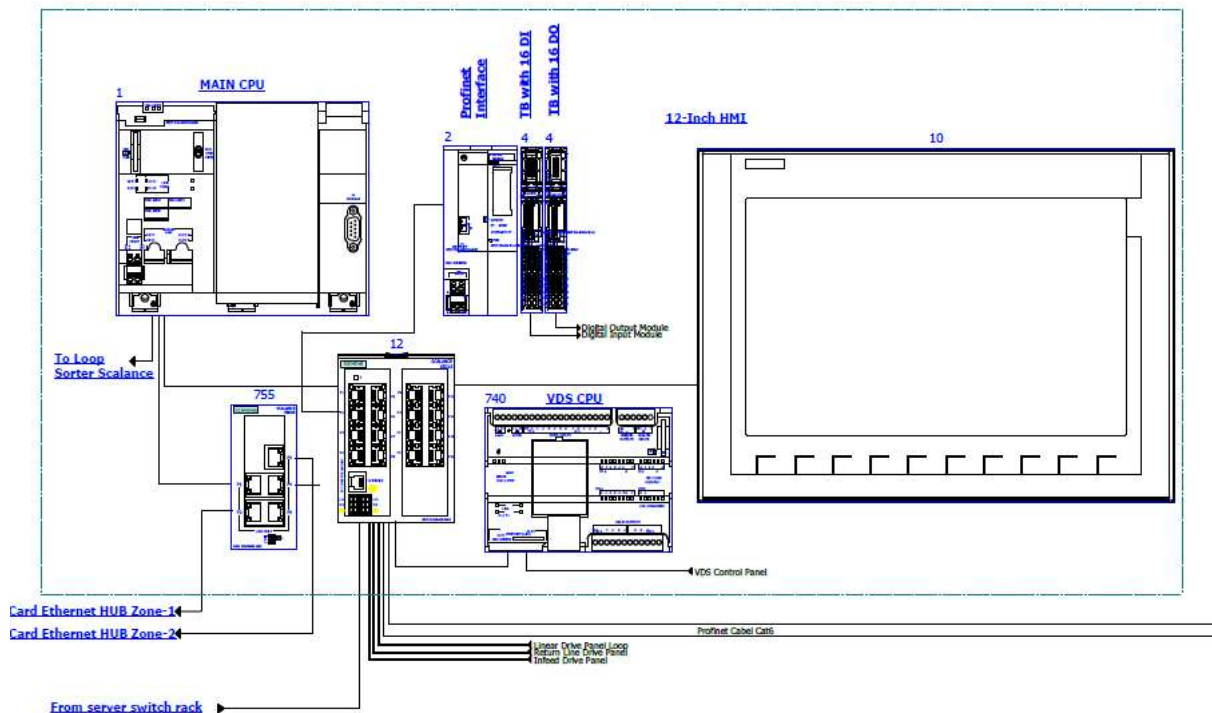
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets. PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

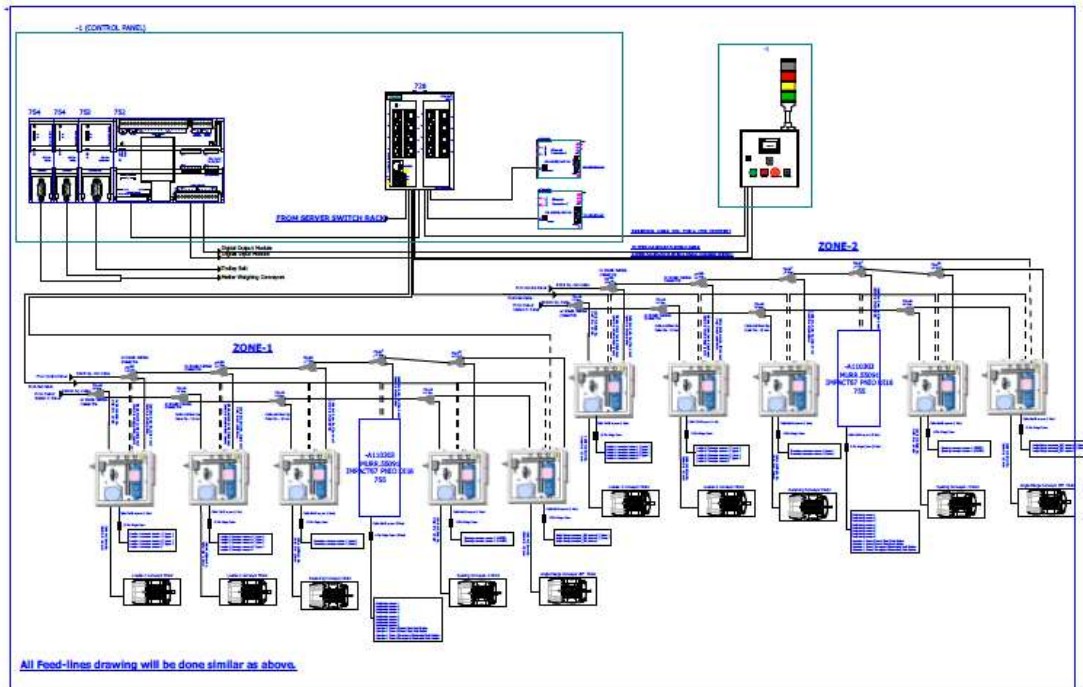
15.1 Reference Picture of Power Distribution Panel



15.2 Main Control Panel (Reference)



15.3 Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter. The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.

All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

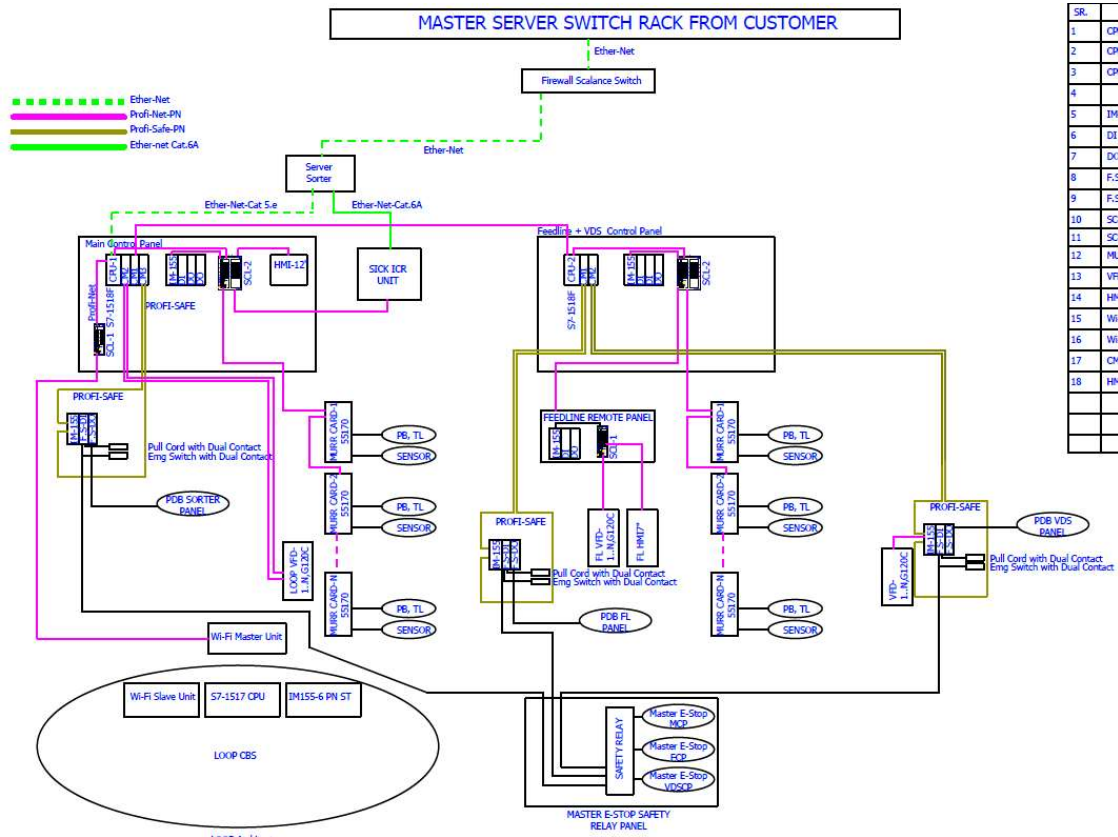
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

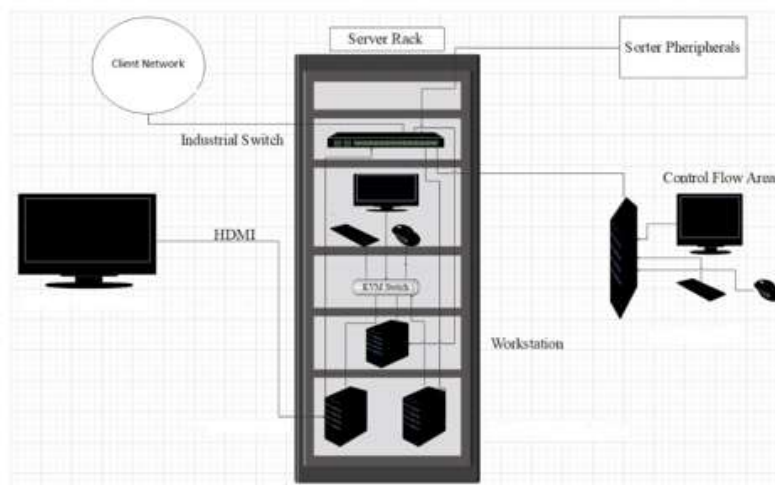
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

16. Controls Architecture (Ref.)



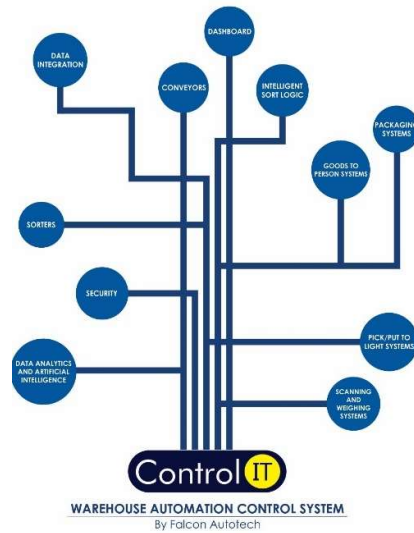
17.IT Architecture (Ref.)



17.1 HAA Server

S.no	Description	Qty
20Core Config with 128 GB RAM in T440 and 64 GB RAM in T40		
1	Synology_storage_DS723+ 2.6 GHz AMD Ryzen R1600 Dual-Core, 2 x Gigabit Ethernet Ports,	2
	2GB ECC DDR4 RAM, 2 x 3.5/2.5" Bays 2 x M.2 2280 Slots E10G22-T1-Mini 10GbE RJ-45 network upgrade module for compact Synology servers	
2	Dell Tower Model T440 -PowerEdge T440 Xeon Gold 6148 2.4GHz/20C/27.5MB/150 W 16 DIMMS 4 x32GB RDIMM Up to 8 3.5"" Hot Plug Hard Drives Tower Configuration 2 x 1.2TB 10K RPM SAS 12Gbps 512n 2.5in	2
	Hot-plug Hard Drive PERC H750 Adapter Full Height 8GB Cache Dual Hot-plug PS495W iDRAC9Enterprise 3YR ProSupport Next Business Day Onsite/Dual Port Lan	
3	Dell PowerEdge T40 Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 4*16 GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3	1
	Year Onsite NBD,480 GB SSD,1 Gbe Dual Lan Card Extra	
4	Windows Server 2019	3
5	Monitor	1
6	KVM Switch	1
7	Mouse	1
8	Keyboard	1
9	Lan Cable	10
10	Power Cable	8
11	VGA Cable	1
12	42 AC Server Rack	1
13	Netgear 24 Port Giga Switch	2
14	2 TB SSD Micron	4
15	DP to VGA Convertor Cadyce Brand	1

18. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	Customer Server	FIREBASE	Any Customer Protocol

19. Falcon's Visual Inspection System (SCADA)

SCADA stands for Supervisory Control and Data Acquisition. It is a system of hardware and software components that allows for remote monitoring, control, and data acquisition of industrial processes or facilities

The Visualization system provided by FALCON (or SCADA) allows the monitoring and control of the different systems delivered for the DHL Riyadh Hub. This SCADA system receives from each monitored sub-system all information on their operating status in real time.

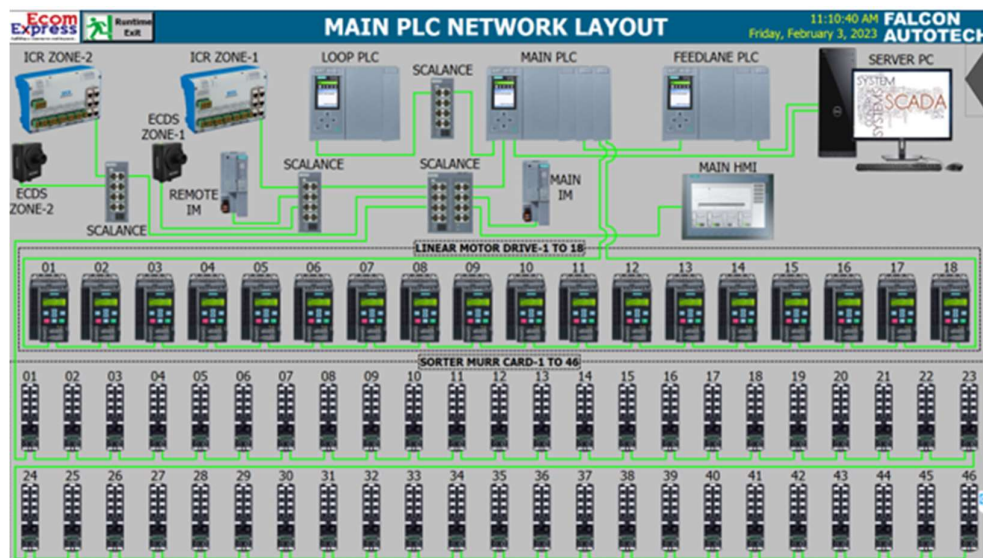
At the system monitoring level, the functions performed are:

- Field data acquisition.
- Animated visualization of equipment.
- Representation of the operating mode of the system (nominal, contingency, etc.).
- Alarm management.
- Alarm history management.
- Diagnostic help.
- Failure detection.
- Equipment control.
- Statistics on equipment operation.
- Historical Statistical Report.
- Recording and archiving.
- Safety operator interface.

19.1 FIELD DATA ACQUISITION

The field data acquisition function is performed by the SCADA system connected to the sorters' PLCs.

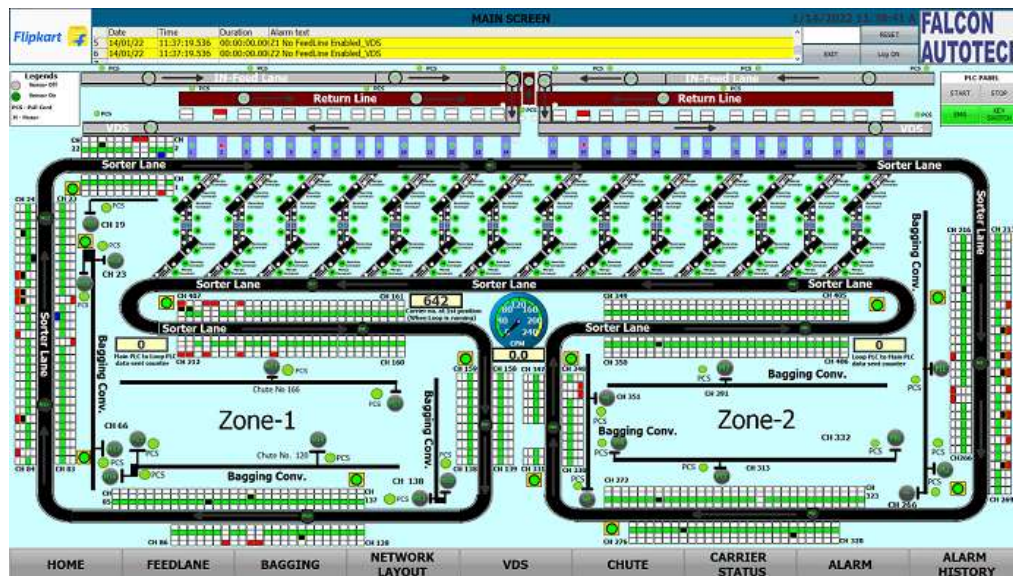
The communication with the PLCs is done using equipped CPU cards that are able to manage the communication with the PLC on the Industrial Ethernet network, without overloading the server. The acquisition of signals from external systems is carried out via the PLC, using dry contacts.



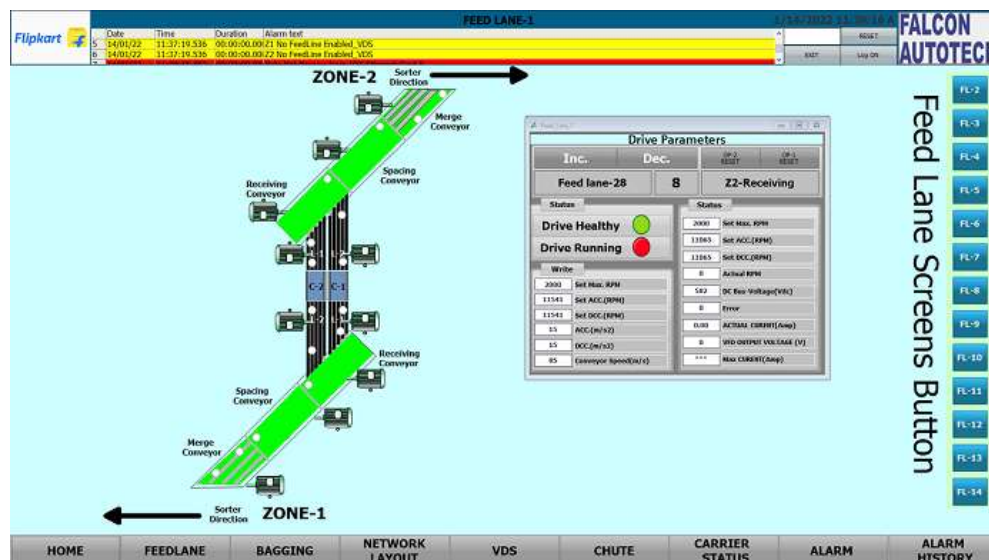
19.2 ANIMATED SYSTEM VISUALIZATION

The animated view represents the dynamic graphical user interface that allows real-time monitoring of the controlled systems and the execution of their control procedures.

The various states of the digital signals are displayed on the screen by graphic symbols. All views are web based with responsive capabilities when required so that various devices can be used to display status and information on PC. The types of information that can be displayed on each type of device will be discussed during the design phase of the project.



Example of a synoptic view



19.3 ALARM MANAGEMENT

The alarm pages display a series of information to identify the nature of the alarm or event, the elements involved and the time.

The alarm management includes:

- The Alarm name.
- The date on which the above-mentioned alarm was activated.
- The moment the alarm goes off.
- The date on which the alarm is activated again.
- The moment the alarm is activated again.
- The sub-system that activated the alarm.
- The area where the alarm was triggered.
- The name of the beacon that activated the alarm.
- The alarm state.
- The alarm value.
- A complete alarm description.

The historic data of each alarm is stored in the SCADA database. This type of fault detected includes:

- ✓ Sensors.
- ✓ Drive Fault.
- ✓ Lack of monitored equipment.
- ✓ Etc.

Alarm Class	No.	Time	Date	Status	Alarm Name
BAG SORTER_ALARM	3265	11:07:24 AM	2/3/2023	1	BAG SORTER-1 ANGULAR FL HAND SCANNER SOCKET ERROR--
FEEDLINE	1918	11:07:43 AM	2/3/2023	1	Feedline Stop Due to Power Save Mode
FEEDLINE	1897	11:07:43 AM	2/3/2023	1	FL_OKT_TCP_Error
FEEDLINE	1898	11:07:43 AM	2/3/2023	1	FL_TCP_Disconnect_Error
FEEDLINE	1672	11:07:43 AM	2/3/2023	1	Feedline Stop Due to Power Save Mode
FEEDLINE	1626	11:07:43 AM	2/3/2023	1	Key Switch Off Error
FEEDLINE	1549	11:07:43 AM	2/3/2023	1	Feedline Stop Due to Power Save Mode
FEEDLINE	1528	11:07:43 AM	2/3/2023	1	FL_OKT_TCP_Error
FEEDLINE	1529	11:07:43 AM	2/3/2023	1	FL_TCP_Disconnect_Error
FEEDLINE	1426	11:07:43 AM	2/3/2023	1	Feedline Stop Due to Power Save Mode
FEEDLINE	1405	11:07:43 AM	2/3/2023	1	FL_OKT_TCP_Error
FEEDLINE	1406	11:07:43 AM	2/3/2023	1	FL_TCP_Disconnect_Error
FEEDLINE	1303	11:07:43 AM	2/3/2023	1	Feedline Stop Due to Power Save Mode
FEEDLINE	1180	11:07:43 AM	2/3/2023	1	Feedline Stop Due to Power Save Mode
FEEDLINE	1134	11:07:43 AM	2/3/2023	1	Key Switch Off Error
BAGGING_ALARM	3829	11:07:24 AM	2/3/2023	1	Bagging Conveyor_MurrCardNode 101 Error_Zone2...

Sample Alarm History Screen

20. Key Components Make

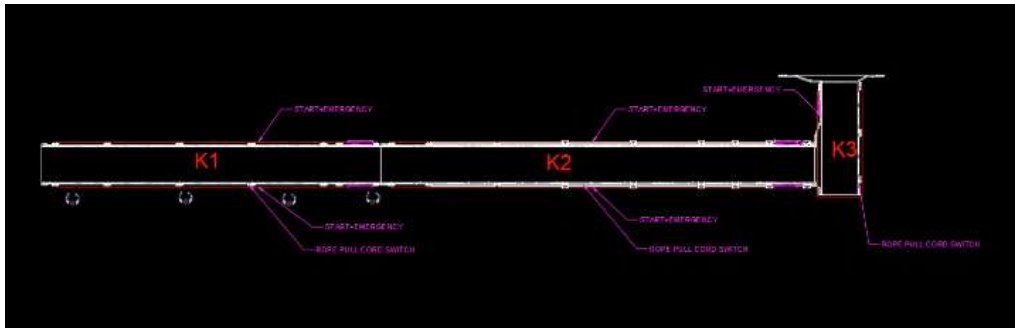
Items	Make
Belts	Forbo/ Derko
Rollers	Falcon
Cross Belt Carriers	Falcon
LIM/LSM	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Volume Scanners (VMS 4200)	SICK
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens/ Omron
Control Panels	Rittal/ BCH
VFDs	Siemens/Lenze/ AB/ Omron
Cables	LAPP/ Equivalent
Switch Gear	Schenider/Equivalent
Bearings	NTN/SKF/Equivalent
Power Transmission Systems	Vahle
HMIs	Siemens

21. Principle of Safety

21.1 E-Stops

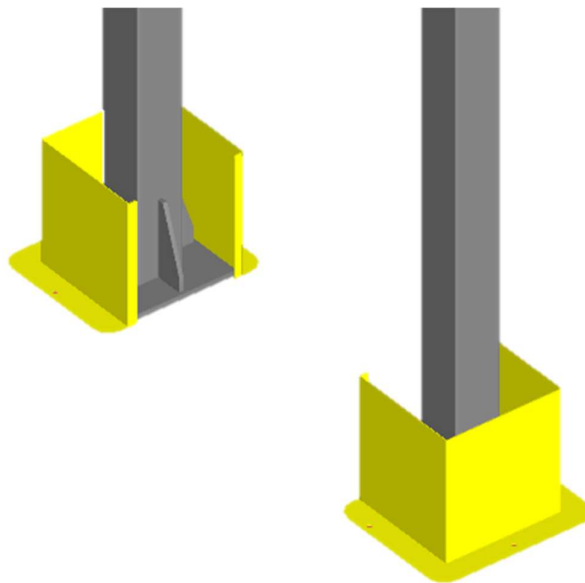


- a. At Every Conveyor Module – Both sides



21.2 Leg Guards:

Leg guards are protective components designed to shield the legs or supports that stabilize machinery. They are essential for preventing accidents, injuries, and equipment damage by covering exposed areas where individuals might come into contact with moving parts or sharp edges. The material of construction is MS.

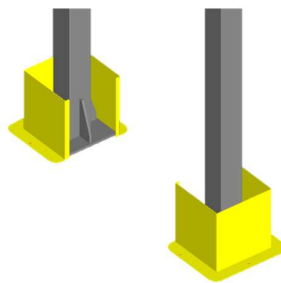


21.2.1 Leg Guard Types:

SL No	Leg Guards	Application	Drawings	Reference Image
1	SPECIAL LEGS	Single Decker CBS Sorter & Special leg structure area for conveyors		
2	XL	DOBLE DECKER / XL CBS		
3	MODULAR LEGS	For Modularization Leg (Standard legs for Conveyors & Chute)		

21.2.2 Leg Guard Rules:

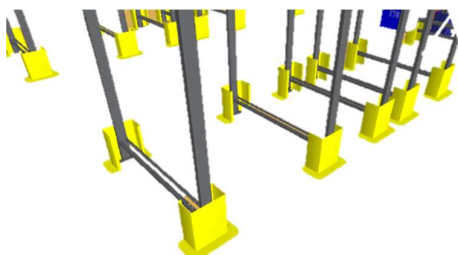
- i. Single Deck/Double Deck and XL CBS: We include leg guards on all CBS legs except in the chute areas.



- ii. **Special Leg Structure for Conveyors:** Where a 2.4m or wider aisle is required underneath the conveyor, we include leg guards for those legs.



- iii. Standard Legs for Conveyors & Chutes: We include leg guards for all conveyors that are running at a height greater than 1.5m.



Any additional Leg guard requirement from client resulting from layout change for example conveyor location, no of pathways or conveyor heights during DAP, should be catered through Change Management process.

22. Program Organisation

22.1 Program/ Project Schedule

S.No.	Activity	Timeline
1	Receipt of PO	Week 0
2	Project Kick off	Week 1
3	DAP Submission	Week 4
4	DAP Approval	Week 5
5	Detailed Designing	Week 8
6	Manufacturing & Procurement	Week 15
7	Inhouse Assembly	Week 17
8	Dispatch Start	Week 19
9	Installation start	Week 21
10	Go Live	Week 26

**Above plan is tentative, detailed plan will be shared post order confirmation.*

22.2 Program & Account Management

At Falcon Autotech, our commitment revolves around fostering optimal engagement and transparency throughout the project management process. To achieve this objective, we propose the implementation of the following approach:

1. Establishment of a dedicated DHL Business Unit within Falcon Autotech, overseeing project management with assigned spokespersons at each hierarchical level to enhance coordination.
2. Implementation of weekly or fortnightly project routines to facilitate meticulous tracking and effective management.
3. Adoption of a robust framework for Change Risk and Opportunity Management, coupled with a governance structure to ensure proactive handling of challenges and opportunities.

23. DHL Responsibility

23.1 DHL Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- DHL to provide HPT/ Forklift for material movement for complete duration.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the shipmentsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

23.2 Responsibilities of DHL During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the shipments correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

23.3 DHL Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of

the course.

- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough space for desks or tables and chairs for the trainer and trained staff.
- Check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technical staff.
- The invitation of staff to attend the training courses will be at the expense of DHL.

24. Testing, Training, and Support

24.1 Acceptance Testing

This section serves to outline the precise criteria and methodology for Falcon to validate that the installed system complies with the operational and performance requirements outlined in this proposal.

Testing necessitates comprehensive assistance from both Falcon and DHL Personnel, including:

- Provision of DHL material/products for functional testing (Falcon requires a sufficient quantity and a complete range of sizes for thorough testing).
- Accessibility to software components and applications (including accurate data for testing).
- Engagement of DHL labour (and equipment if necessary) to facilitate the testing process.
- Ensuring the availability of system equipment throughout the duration of each test.

24.1.1 Acceptance Test Plan

Based on the criteria outlined and mutually agreed upon by Falcon and DHL in this proposal, Falcon will formulate the Acceptance Test Plan (ATP). The ATP will define a systematic approach for the execution of the Acceptance Tests specified in this section. Furthermore, the ATP serves as the instrument employed by Falcon to achieve formal acceptance and obtain sign-off from DHL.

Falcon will furnish the ATP to DHL before the initiation of any tests. The ATP will encompass the following elements:

- Test prerequisites
- Daily plan/schedule
- Personnel responsibilities
- Computer data and test loads required
- Specific test procedures
- Anticipated test results

24.1.2 Minor / Major Faults

Falcon will classify any encountered faults during testing into either Minor or Major categories, as outlined below:

- Minor Faults are characterized as issues affecting a confined area or a single component with no repercussions on the overall test. These faults will be included in the System Snaglist and will not impede the ongoing testing process.
- Major Faults are defined as issues with a significant impact, leading to the inability to showcase the functionality in question. Addressing Major Faults may necessitate restarting the test.

24.1.2.1 Specific Procedure

- Falcon staff will oversee the upkeep and dissemination of the System Snaglist during Acceptance Testing.
- Upon the resolution of each issue, DHL will validate the correction and endorse the resolution on the System Snaglist.

Note: Validation can be conducted independently of Falcon personnel presence and should be documented through the designated signature of DHL on the System Snaglist.

24.1.3 Included Acceptance Tests

The following Acceptance Tests are part of the Final Acceptance Testing:

- Physical Acceptance Testing.
- Functional Acceptance Testing.
- System throughput Testing (as specified).

24.1.4 Physical Acceptance Testing

24.1.4.1 Test Definition

The Physical Test involves a walkthrough of the System, attended by both Falcon and DHL personnel. They will verify that the physical equipment is complete and installed according to the final drawings and specifications, typically documented by area or subsystem.

24.1.4.2 Specific Practice

- Physical Testing is conducted continuously and concurrently with installation.
- Physical Testing may extend into the commissioning and integration periods and should be completed before other Acceptance Testing.
- Physical Testing is considered complete after all System walkthroughs and the resolution of items recorded in the System Snaglist.

24.1.5 Functional Acceptance Testing

24.1.5.1 Test Definition

Functional Testing follows the completion of Falcon engineering commissioning activities. It verifies the proper function of the System according to the requirements in this Proposal and the detailed specifications developed during the Project.

Note: This testing does not repeat each test conducted during the Commissioning phase of the Project.

Both Falcon and DHL personnel will attend and verify that the System functions as described in this Proposal. Defects requiring correction will be logged into the System Snaglist.

Functional Testing typically includes:

- Operation/equipment interface, safety mechanisms, and interaction testing.
- Electrical/control functions such as normal start/stop, flow control, and emergency stopping.
- Controller and computer system interfaces (uploads, downloads, and interlock signals).

24.1.5.2 Specific Practice

Functional Testing is considered complete after the successful conclusion of all Functional Tests and the resolution of items recorded in the System Snaglist.

24.1.6 Throughput Testing

24.1.6.1 Test Definition

To demonstrate System's ability to meet the system throughput in the final Material Flow Throughput section of this Proposal, Falcon proposes to conduct Mechanical System Throughput Testing for each area defined in the ATP before beneficial use of the system.

24.1.6.2 Measurement

System Throughput Testing is completed by Falcon under non-production/controlled conditions, after the completion of related commissioning activities. Defects requiring correction are logged into the System Snaglist.

Tests verify that system throughputs are achieved for the areas specified in the ATP.

24.1.6.3 Specific Practice

System Throughput Testing is considered complete after the successful conclusion of all System Throughput Tests.

24.1.7 Final Acceptance Criteria

Final Acceptance is declared complete when:

- Physical Acceptance Testing, Functional Acceptance Testing, and System Throughput Testing are all complete.
- All items in the System Snaglist are resolved.
- As-built documentation is submitted.

24.1.7.1 Acceptance Test Letter

After successful completion of Acceptance Testing, Falcon will provide DHL a letter addressing the following:

- The System has been installed and accepted by DHL.
- Note: Acceptance may be conditional on the remedy of agreed-upon System Snaglist items with established completion dates.
- Confirmation of the commencement of the warranty period.

- The final payment amount required and the designated due date.

Note: Falcon has approached the testing for this project based on Falcon standard testing practices.

24.2 Training

Training for the recently installed material handling system at DHL's facility is an integral part of the implementation process. Falcon employs a business-linked learning approach in the design, development, delivery, and evaluation of its training programs.

Falcon adheres to a well-defined process tailored to create learning content specific to DHL's installation and application of their systems.

24.2.1 Operations Training

System Operations Training is provided to offer participants an understanding of their area from an operational perspective. Falcon targets critical operations in the assigned functional area, covering flow control, system operation, and control devices. Discussions also emphasize the impact of individual and area performance on upstream and downstream workers.

24.2.2 Maintenance Training

The Maintenance Training curriculum encompasses mechanical, electrical, and controls aspects of the installed system. Topics include safety, system construction and installation, system operation, maintenance and repair procedures, and technical documentation.

Falcon recommends assigning personnel responsible for system maintenance to work with the installation and commissioning crews during their final weeks on-site. Formal training courses for maintenance personnel will be conducted before system startup, featuring classroom lectures, audio-visual presentations, and hands-on demonstrations with the installed system. Site tours will be organized to highlight common operational issues affecting system uptime.

24.2.3 Information System Training

Information System training covers the computer hardware and software aspects of the Automated Material Handling System. Software Application Engineering topics include system administration, data storage, computer system operation, and other computer-related tasks.

25. Life Cycle Services

Falcon can provide as an option the following Life Cycle services:

- Preventive maintenance.
- Corrective maintenance.
- Hotline service.
- On Site Support
- Supply and management of spare parts.

25.1 Preventive Maintenance

A preventive maintenance package refers to planned maintenance activities and services designed to prevent equipment, machinery, or systems from breaking down or deteriorating over time. Preventive maintenance aims to identify and address potential issues before they lead to costly breakdowns or disruptions in operations.

25.2 Corrective Maintenance

Corrective maintenance, also known as corrective maintenance, is a type of maintenance activity that focuses on addressing and repairing equipment or system failures, defects, or malfunctions when they occur. Unlike preventive maintenance, which aims to prevent breakdowns through scheduled inspections and maintenance tasks, Corrective maintenance is reactive and comes into play when a problem has already arisen.

25.3 Hotline Support

The hotline service access is available 24 X 7

The main function is to dispatch the call to the proper qualified engineer for immediate answer. If the Falcon qualified engineer is not available, the call is transferred to a back-up qualified engineer and finally an answering machine. In the last case, the DHL personnel shall be contacted within the time frame described in the service level.

The Falcon qualified engineer will support DHL maintenance personnel by using a remote access (via VPN) to the equipment/systems.

25.4 On Site Support

To adhere to the committed System uptime, dedicated On site team will be available at Site covering all shifts. Team comprises of Supervisor in general shift and 1 Electrical & 1 Mechanical technician in each Shift.

25.5 Supply and Management of Spare Parts

This paragraph describes the principles and procedures that will be implemented between Falcon and DHL, with regard to the spare parts management policy.

- Spare parts orders.
- On completion of the DAP, when the configuration will be finalized a final and detailed list, which will be extended to include all spare and wear parts, will be presented to DHL.
- The exact number of spare parts will be determined with DHL, according to its stock strategy on the basis of the list of recommended spare parts.

26. Commercial Terms

26.1 Price Sheet

Sr. No	Component	Set	UOM	Total(INR)
1	Loop Cross Belt Sorter with VMS, ICR	1	Set	₹ 4,11,42,915
2	Conveyors	1	Set	₹ 47,22,049
3	Destinations	1	Set	₹ 70,89,139
4	Steelwork	1	Set	₹ 57,37,500
5	Software Packages	1	Set	₹ 6,00,000
6	Installation & Commissioning	1	Set	₹ 29,34,580
7	Packaging & Forwarding	1	Set	₹ 11,73,832
8	Project Management and Engineering Charges	1	Set	₹ 15,85,000
Grand Total				₹ 6,49,85,015

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Freight charges shall be extra at actual.
- Taxes : GST @18%
- Including Packing and Forwarding
- Duty unpaid
- Installation and Commissioning: Included.
- Price is valid for 30 days from the date of the proposal.

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. The same shall be borne & paid by client itself.

**Material Unloading & Laydown area is in customer scope*

The above price is budgetary figures & will change when actual system derived costing is done.

26.2 Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 50% Advance along with PO.

Stage 2: 40% + Taxes after Delivery within 15 days.

Stage 3: 10% after successful Installation within 15 Days.

27. Warranty Period

Falcon's offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use).

The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)

The warranty does not apply to the replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

28. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Mobile Carts**
- **Construction Power**
- **Steel Works- If not specified**
- **Collection Trolleys below chutes**
- **PC for SCADA.**
- **Fans at chutes & Inducts.**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices.
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.